

Is research on hedge fund performance published selectively?

A quantitative survey

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Abstract

We examine whether estimates of hedge fund performance reported in prior empirical research are affected by the publication selection bias. Using a sample of 1,019 intercept terms from regressions of hedge fund returns on risk factors (the “alphas”) collected from 74 studies published between 2001 and 2021, we show that inferences about hedge fund returns are not significantly contaminated by selective publication of empirical results. Most of our monthly alpha estimates adjusted for the (small) bias fall within a relatively narrow range of 30 to 40 basis points. Considering several partitions of our sample, we document a modest publication bias only for estimates based on IV, for which relatively large standard errors are common, and that tend to be less precise. In contrast, studies that explicitly control for the potential biases in the underlying data (e.g., the backfilling bias and the survivorship bias) report lower alphas. Our results demonstrate that despite the prevalence of the publication selection bias in numerous other research settings, publication may not be selective when there is no strong *a priori* theoretical prediction about the sign of estimated coefficients, which may induce greater readiness to publish statistically insignificant results.

Keywords: Hedge funds, meta-analysis, publication bias

JEL Codes: J23, J24, J31

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1 Introduction

“I can’t figure out why anyone invests in active management, so asking me about hedge funds is just an extreme version of the same question. Since I think everything is appropriately priced, my advice would be to avoid high fees. So you can forget about hedge funds.”

Eugene F. Fama ¹

Over the past three decades hedge funds have experienced a spectacular increase in popularity. The value of assets under management (AUM) increased about 100 times between 1990 and 2020 (Stulz, 2007; Barth *et al.*, 2020). This trend is difficult to reconcile with the Efficient Market Hypotheses (EMH) (Fama, 1970), which suggests that financial markets rationally process available information and establish unbiased pricing of traded assets. Efficient financial markets should quickly eliminate any opportunities to earn abnormal return. In efficient markets, an optimal investment strategy involves passive holding of a broad portfolio of assets (i.e., “indexing”). Consistent with this proposition, there has been a sharp rise in passive investment in recent decades. Gârleanu & Pedersen (2022) report that the proportion of wealth invested in passive funds increased from close to zero in 1990 to over 15% in 2017. Easley *et al.* (2021) argue that this trend has been further accelerated by the arrival of exchange-traded funds (ETFs). As passive funds do not participate in information processing and price discovery, the rise of passive investment raised concerns about its potential detrimental impact on market efficiency. However, as argued by Stambaugh (2014), any deviations from efficient pricing create new opportunities for active investment. Thus, even though the proportion of actively managed funds decreased over time (Stambaugh, 2014), active mutual funds and hedge funds still manage more than 30% of invested wealth, which is more than twice as much as passive mutual funds and ETFs (Gârleanu & Pedersen, 2022).

As aptly expressed in the opening quote by Eugene F. Fama, the 2013 Nobel laureate in economics, hedge funds represent an extreme form of active investment management. Consequently, they charge investors high management and performance fees. Ben-David *et al.* (2020) estimate that for every dollar of gross excess return earned by a hedge fund, on average, 64

¹Source: <https://www.azquotes.com/quotes/topics/hedge-fund.html>.

cents are paid in management and incentive fees and only 36 cents are collected by the investors. Given the magnitude of these fees, it is surprising that hedge funds keep attracting growing amounts of capital.

A potential explanation for this puzzling trend is that investors have distorted views of the value hedge funds actually generate. Hedge funds are relatively lightly regulated and so they remain rather opaque in terms of their investment strategies, asset holdings, and realized returns. The paucity of information constrains systematic analysis of hedge fund performance and it may bias investors' expectations about hedge funds' value generating potential. Hedge funds are not obliged to periodically publish information on their performance. Some but not all voluntarily report their performance data to commercial data providers. This implies that data available for research is fragmented and it may suffer from numerous biases. Furthermore, hedge funds tend to engage in a wide range of unconventional investment strategies and so it is not trivial to adequately adjust for the risks they bear. It is not clear to what extent these constraints bias reported performance estimates. Prior empirical literature includes numerous conflicting results, which make it difficult to draw clear conclusions. Research literature thus lacks a study synthesizing this pool of diverse empirical results.

In this paper, we perform the first quantitative survey of research literature on hedge fund performance. Our aim is to review and integrate published empirical findings and to examine how they are affected by publication selection and data biases. Brodeur *et al.* (2020) argue that research methods that offer researchers more degrees of freedom are more likely to suffer from selective publication as researchers may choose research designs and data sample to generate results that are attractive for publication. We argue that the fragmentation of hedge fund performance data and the wide range of alternative approaches to controlling for risk give researchers considerable discretion in research design. Various data sources and estimation techniques may produce different results, some of which may be more attractive for publication than others. This potentially creates opportunities for selective publication that could bias the pool of coefficients reported in research articles. Hence, we believe that research literature on hedge fund performance merits a systematic investigation of the prevalence of this potential bias and of its potential impact on the reported estimates. Nevertheless, to the best of our knowledge, no prior study estimates and corrects for publication bias in this stream of literature. We provide

the first quantification of the impact of potential selective publication and data biases on the hedge fund performance estimates.

We review empirical results in 74 studies on hedge fund performance published between 2001 and 2021. Our analysis is based on a sample of 1,019 estimates of intercept terms (i.e. the “alphas”) from regressions of hedge fund returns on risk factors. The risk factors on the right-hand side of the regression equation represent various risk dimensions, to which hedge fund investments may be exposed to. The slope coefficients (i.e. the “betas”) capture hedge funds’ exposure to the individual risk dimensions. The intercept terms, the alphas, represent the portion of realized returns that is not attributable to the fund’s exposure to the systematic risk factors. In other words, the alphas represent the abnormal return earned by the hedge fund, which represents the difference between the actual realized return and the return that would be expected as fair compensation for the investment’s systematic risk.

We examine the extent to which the alpha estimates are affected by selective publication and data biases. A publication selection bias represents a tendency to publish empirical results that are consistent with the underlying theoretical predictions or with prior empirical findings. Selective publication may result from both conscious and subconscious decisions made by authors, editors, and referees who discard results that look implausible in the light of their *a priori* expectations (Ioannidis *et al.*, 2017). Publication selection bias and its implications are extensively discussed in prior literature including Stanley (2001, 2005); Stanley *et al.* (2010); Havranek (2015); Brodeur *et al.* (2016); Bruns & Ioannidis (2016); Stanley & Doucouliagos (2017); Christensen & Miguel (2018); Brodeur *et al.* (2020); Blanco-Perez & Brodeur (2020); Zigràiova *et al.* (2021). These studies document that publication selection bias is indeed widespread in a wide range of economic settings and it has a substantial impact on the mean value of reported estimates. Given the discretion in research design due to data fragmentation and the variety of risk-adjustment methods, it is worthwhile to examine whether a similar bias is present also in the empirical literature on hedge fund performance.

We use several approaches to test for the publication selection bias. First, we exploit the property that tests of statistical significance typically assume that publication bias is a linear function of the standard error. Hence, documenting a correlation between the two can be used as evidence on biased reporting of results in primary studies (Egger *et al.*, 1997). We

complement this conventional approach with several other recently developed techniques that use different combinations of fixed effects and weighting, that relax the assumptions about the underlying distribution of the estimated coefficients, and exploit discontinuities in these distributions (Stanley *et al.*, 2010; Ioannidis *et al.*, 2017; Bom & Rachinger, 2019; Furukawa, 2020; Andrews & Kasy, 2019; van Aert & van Assen, 2020). Using these modern techniques allows us to evaluate the robustness of our findings to assumptions that underlie various methodological approaches.

We find that despite the multitude of data sources and methodological approaches, empirical research on hedge fund performance is not substantially contaminated by the publication selection bias. In our regressions, most of the slope coefficients that capture the impact of publication bias are statistically insignificant. Furthermore, the monthly alpha estimates adjusted for this (small) bias are fairly close to our unconditional sample mean of 36 basis points (i.e. 0.36%, which corresponds to 4.3% *per annum*). These aggregate results are based on the entire pool of alphas as they are reported in prior research literature on hedge fund performance. They are estimated using various data sets and methodologies. Most of the alphas in our data set are computed on the net-of-fee basis (984 out of 1,019, not tabulated), which implies that the average value we report mostly represents abnormal returns net of management and performance fees. Based on various techniques, we obtain estimates of the “representative” alphas corrected for the publication selection bias ranging from 0.274 and 0.386. We observe fairly similar ranges when only using observations published in the Top 3 (0.265, 0.358) and Top 5 (0.263, 0.355) finance journals, which suggests that publication quality does not materially effect the reported estimates. In comparison, we document a slightly higher ranges for estimates based on the one-factor model (0.349, 0.707) and those that are not treated for the survivorship and backfilling biases (0.282, 0.521). In contrast, we observe a lower and wider range for the “corrected” alpha coefficients based on IV techniques (-0.411, 0.298).

In addition, our results also mostly hold when we consider more homogeneous subsamples of alpha estimates that either adjust or do not adjust for survivorship and backfilling biases, subsamples that use a specific asset pricing model to compute the alphas, and subsamples of alphas sourced from the three and five leading journals in finance. A notable exception is the group of empirical estimates based on IV. Such a conclusion is consistent with a recent paper by

Brodeur *et al.* (2020) who find that IV-based estimates are more likely to suffer from publication bias than estimates based on other techniques. Furthermore, we observe little publication bias in data subsamples partitioned based on whether a given estimate explicitly adjusts for potential biases in the underlying data (e.g. the backfilling bias and the survivorship bias). However, consistent with the expected relevance of these biases discussed in hedge fund performance literature, we observe somewhat lower alphas in studies that adjust for these data biases.

We make several important contributions to the literature. First, using several recently developed methodological approaches we aggregate and synthesize fragmented empirical evidence on hedge fund performance. Prior research has long acknowledged that the absence of a comprehensive database may lead to distortion and misleading findings (Fung & Hsieh, 2004a; Agarwal *et al.*, 2009). Fung & Hsieh (2004a) point out that differences in coverage across various hedge fund data providers may lead to rather different average returns for a given hedge fund type. To illustrate this observation, they state that two data providers specified two substantially different estimates for equity market-neutral hedge funds for the month of January 2001 (-1.57% vs. 2.13%). Such discrepancies across data sources imply that the choice of the database may have a substantial impact on the estimated hedge fund performance. In a recent working paper, Joenvaara *et al.* (2019a) underscore the importance of combining data from various databases and they propose a new way of doing so. We offer an alternative approach to overcome the data fragmentation problem. Our approach is based on aggregating the alpha coefficients estimated in prior studies that are themselves based on various data sources. Relative to Joenvaara *et al.* (2019a) the advantage of our approach is that it allows us to include even estimates based on private or hand-collected data and to simultaneously control for potential data and publication biases.

Second, we provide a comprehensive battery of tests to evaluate the impact of the publication selection and data biases on hedge fund performance estimates. This paper is the first study that systematically analyzes the impact of selective publication on the reported hedge fund performance results. Recent research suggests that research settings that offer researchers considerable discretion are particularly prone to suffer from selective publication (Brodeur *et al.*, 2020). We evaluate this effect in a research field that is characterized by fragmented data and a plurality of methodological approaches to estimate abnormal returns. Furthermore, hedge fund

literature frequently mentions a concern that survivorship and backfilling biases may distort estimates of hedge fund performance (Fung & Hsieh, 2004a). Prior studies typically address this issue by measuring the returns of funds of funds as their reported performance is less likely to be affected by backfilling historical information for successfully incubated funds and by omitting data for dead funds. However, relying on the data on funds of funds has several shortcomings. First, the inclusion of a hedge fund in a fund of funds is in itself an endogenous decision that has an impact on the reported returns. There is no guarantee that the funds that are actually included in the fund of funds' portfolios are representative of the entire hedge fund population and that the individual funds are treated in these portfolios with appropriate weights relative to the general population. Second, funds of funds charge investors an additional layer of management and performance fees (Stulz, 2007), which may distort the quantification of the abnormal return generated by individual hedge funds (Amin & Kat, 2003). Due to these additional costs, funds of funds may not represent an attractive investment opportunity for many hedge fund investors. We offer a different approach to adjust for these issues that is based on the aggregation of estimates reported in prior studies.

Third, by documenting a publication selection bias for the subset of estimates based on IV our study provides out-of-sample evidence in support of the recent finding by Brodeur *et al.* (2020) who argue that IV-based estimates are more likely to suffer from publication bias than estimates based on other techniques. When exploring the potential underlying reasons for this finding, Brodeur *et al.* (2020) suggest that it may arise due to the considerable discretion IV estimation gives researchers in designing their empirical tests. In line with this conjecture the authors observe that when the instruments are relatively weak, the second stage results are likely to be close to the conventional thresholds for statistical significance. Our evidence is consistent with this proposed explanation. We observe that IV-based estimates in our sample seem to be more likely to suffer from selective publication.

Our analysis is relevant to investors who consider investing in hedge funds, to regulators who seek the optimal design of the regulatory framework, as well as to researchers in economics and finance. Our evidence on the absence of a significant publication bias and the fairly narrow range of 30 to 40 basis points that we document for the corrected monthly alpha estimates allow investors to calibrate their expectations of hedge fund performance. Our study also

informs regulators that even though hedge funds are not obliged to systematically publish their performance and the data is fragmented in numerous private databases, prior empirical research does not suffer from selectivity in reporting hedge fund performance. Finally, our study demonstrates that despite the prevalence of the publication selection bias in numerous other research settings in economics and finance, publication tends not to be selective when there is no strong *a priori* theoretical prediction about the sign of the estimated coefficient and when journals may be more open to publishing statistically insignificant estimates. This may help researchers identify areas where the publication bias can be expected and where, in contrast, it is less likely.

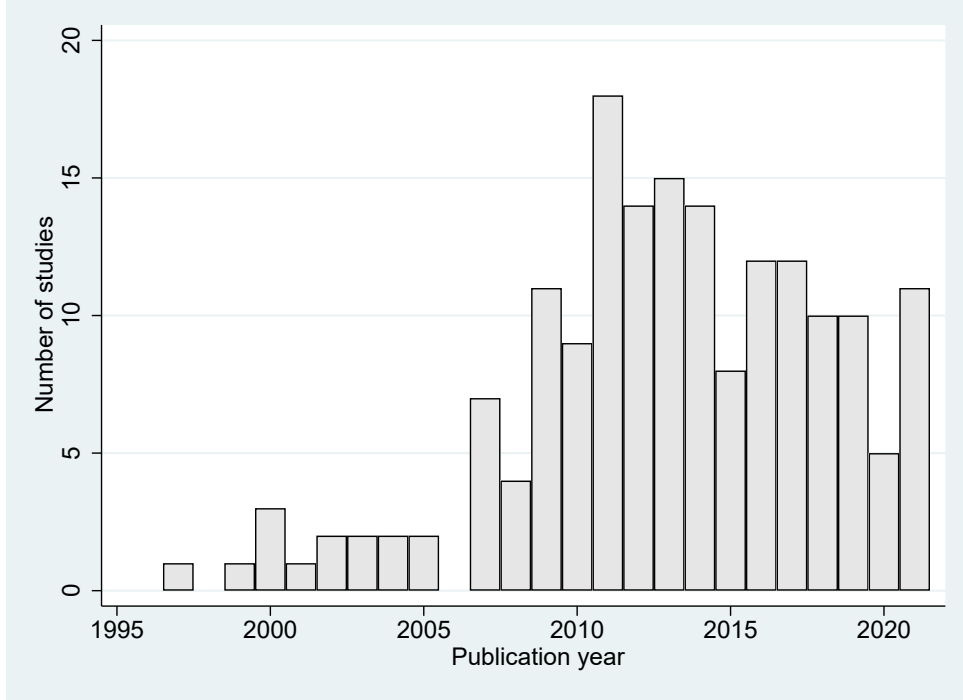
The remainder of the paper is organized as follows. In Section 2, we review prior research literature. In Section 3, we describe our data collection procedure. In Section 4, we present our main empirical results based on the full sample of alpha estimates collected from the primary studies. In Section 5, we report the results based on more homogenous subsamples of alpha estimates. Section 6 concludes.

2 Literature

The increasing prominence of hedge funds as an investment device and the increasing role they play in the economy prompted extensive empirical research aimed at evaluating how well they perform. Over the past decade numerous studies on hedge fund performance have been published. Figure 1 shows the surging number of studies on hedge funds published in leading finance journals – the *Journal of Finance*, the *Journal of Financial Economics*, the *Review of Financial Studies*, the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*.

Connor & Woo (2004) and Agarwal *et al.* (2015) provide narrative reviews of the hedge fund literature. Connor & Woo (2004) give an overview of the history of hedge funds, they discuss the key characteristics that distinguish hedge funds from other investment vehicles, outline their typical investment strategies, and discuss issues in measuring hedge fund risk and performance. Agarwal *et al.* (2015) concentrate more specifically on reviewing research on hedge fund performance and on factors that affect it, e.g., hedge fund characteristics and the risks hedge funds take. They also discuss the role hedge funds play in the economy and their

Figure 1: Articles on hedge fund performance



Note: The figure shows the number of hedge fund-related articles in top 5 journals in finance (*Journal of Finance*, *Journal of Financial Economics*, *Review of Financial Studies*, *Review of Finance*, *Journal of Financial and Quantitative Analysis*) published in a given year excluding articles that are only published online without a print version.

impact on asset prices, market liquidity, quality of corporate governance, and the propagation of financial crises. Furthermore, they discuss a number of issues related to the sources of data used for hedge fund research. We complement these studies by performing a quantitative analysis of whether estimates of hedge fund performance reported in prior empirical research are affected by the publication selection bias.

2.1 Estimating Performance

A standard challenge addressed in empirical research analyzing the performance of investment strategies (including those followed by hedge funds) is to properly adjust for the systematic risk these strategies involve. To address this issue, most modern studies on hedge fund performance report the intercept terms (the “alphas”) from regressions of hedge fund returns on various combinations of risk factors, as shown in Equation 1.

$$(R_p - R_f) = \alpha_p + \sum_{n=1}^N \beta_{n,p} \cdot F_n + \epsilon_p \quad (1)$$

where R_p denotes the realized return on portfolio p , R_f denotes the risk-free rate of return, α_p represents the intercept term, F_n represents the n -th risk factor, $\beta_{n,p}$ denotes the sensitivity of portfolio p to the n -th risk factor, and ϵ_p represents the error term. Loadings on the risk factors (the “betas”) represent a “normal” compensation for the risk that the investment entails the alphas capture the portion of realized returns unexplained by the set of risk factors. The alphas can thus be interpreted as “abnormal” returns that the fund generates for the investors over and above (or below) what would be expected for a given level of risk. This approach explicitly models an investment’s exposure to various risk dimensions. However, the set of relevant risk dimensions is open to question. Thus prior literature provides estimates based on various risk models.

The Jensen (1968) alpha is the simplest of the intercept-based approaches. It was initially designed to measure the investment performance of mutual funds. Returns are measured relative to a benchmark that is relevant for a well-diversified investor. Building on the portfolio theory (Markowitz, 1952) and the Capital Asset Pricing Model (CAPM) (Sharpe, 1966; Lintner, 1965; Mossin, 1966; Black, 1972) this approach uses the equity market excess return ($R_m - R_f$) as the sole risk factor. It maintains that well-diversified investors only require compensation for an investment’s contribution to the volatility of returns on the market portfolio, i.e. for an investment’s systematic risk, which is in turn determined by its returns’ sensitivity to the variation in market returns. The slope coefficient beta in a regression of an investment’s excess return on the market portfolio excess returns captures this sensitivity. In contrast, the intercept term alpha represents the portion of realized excess return that cannot be explained by an investment’s contribution to the portfolio risk, i.e. the value generated for investors.

The simplicity in modeling systematic risk with the use of a single risk dimension constitutes a limitation that may be particularly relevant for hedge funds that engage in complex and dynamic investment strategies that are likely to exhibit various forms of exposure to systematic risk. Due to this complexity, prior research develops risk models that are specifically designed to capture risk dimensions relevant to hedge fund strategies. Most prominently, Fung & Hsieh (2004a); Fung *et al.* (2008) propose a seven-factor model that comprises risk factors that mirror

various risk exposures common in popular hedge fund investment strategies. Specifically, the model comprises the following risk factors: (i) the stock market excess return, (ii) the spread between the small-capitalization and large-capitalization stock returns, the excess return pairs of look-back call and put options (iii) on currency futures, (iv) on commodity futures, and (v) on bond futures, (vi) the duration-adjusted change in the yield spread of the U.S. 10-year Treasury bond over the 3-month T-bill, and (vii) the duration-adjusted change in the credit spread of Moody’s BAA bond over the 10-year Treasury bond. These risk factors are intended to capture risk exposures of a broad set of hedge fund types ranging from equity long-short funds to managed futures funds.

The use of various pricing models has some advantages and disadvantages. The Jensen alpha is well-rooted in financial theory and universally applicable in a wide range of research settings. Hence, empirical results based on the Jensen alpha are easily comparable across various research strategies and data samples. Furthermore, the Jensen alpha is not directly affected by the “model uncertainty problem”, which results from the uncertainty over which of the multitude of variables identified in prior research to be associated with realized stock returns actually constitute the “true” returns determinants. Harvey *et al.* (2016) and Harvey (2017) suggest that the pool of candidate risk factors documented in prior research is overblown by the tendency to publish statistically significant rather than insignificant results, i.e., “*p*-hacking”². Thus, the authors argue that the variables associated with realized returns may not be true risk factors. Instead, they may represent “false positives” that will likely fail to explain the cross-sectional variation in realized returns in the future. In contrast to the universal measure of the Jensen alpha, the seven-factor model is specifically designed for research in hedge fund performance. Thus, it is likely more effective in filtering away various risk exposures relevant to complex investment strategies followed by hedge funds. Prior research offers alpha based on various risk models. In our robustness checks, we evaluate whether our findings are sensitive to limiting our analysis to a subsample of alpha coefficients that are based on the one-factor and seven-factor models.

²We explicitly address the issue of “*p*-hacking” in our research setting in subsection 4.2

2.2 Data Fragmentation

Besides the uncertainty about the appropriate risk model, hedge fund performance research faces the challenge of data fragmentation. Due to the relatively light regulatory oversight, hedge funds are mostly not obliged to periodically disclose audited financial information on their performance. Hence, there is no comprehensive central depository of hedge fund data. Only a subset of funds self-select to voluntarily report information on their performance to private data providers. Prior research thus mostly relies on data sets obtained either from commercial databases or hand-collected. The data sets used in prior research may not be comprehensive and so they may not be fully representative of the entire hedge fund population (Aggarwal & Jorion, 2010; Liang, 2000; Posthuma & Van der Sluis, 2003). This may complicate the interpretation of these findings and raise questions about the generalizability of these results to the universe of hedge funds.

Fung *et al.* (2006) discuss the level of overlap in hedge fund coverage between various databases. Liang (2000) and Agarwal *et al.* (2009) show that the information provided is not always consistent across all the databases, which implies that the results reported in prior research may be sensitive to the choice of the source database. Similarly, in a recent working paper Joenvaara *et al.* (2019a) propose a new way of combining data from various databases and they conclude that using this combined database matters for conclusion about hedge fund performance. They argue that based on this combined database hedge fund performance appears to be lower but more persistent. These findings underscore the importance of aggregating results based on different segments of hedge fund data.

Hedge funds are not obliged to independently verify reported data by auditors or established data providers. Liang (2003) finds that surviving funds are more likely to be effectively audited and funds with more reputable auditors report more consistent data. Patton *et al.* (2015) find that data on hedge fund returns change depending on when the database is accessed. They also observe that underperforming funds are more likely to alter their performance histories. Data on hedge fund returns may be unreliable because valuation of illiquid holdings may be imprecise (Cassar & Gerakos, 2011) or because the highly incentivized managers may tamper with the reported information to give an impression of better and more stable performance (Bollen & Pool, 2009). These complications may contaminate the results of hedge fund research and affect

inferences about overall hedge fund performance.

2.3 Empirical Findings

Given the multitude of data sources and the range of methodological approaches used to estimate hedge fund performance, it is not surprising that prior research amassed an extensive body of sometimes conflicting empirical findings. Several studies indicate that hedge funds generate value for investors. Brown *et al.* (1999) document superior risk-adjusted returns in offshore hedge funds, but they find little support for performance persistence. Ackermann *et al.* (1999) and Liang (1999) observe that hedge funds earn higher risk-adjusted returns than mutual funds even though they have a higher overall risk due to which hedge funds do not outperform general stock market indices. Agarwal & Naik (2000) find that combining investments in hedge funds with passive investing generates better reward-risk combinations than a passive investment in various asset classes. Fung & Hsieh (2004a) propose seven risk factors relevant to hedge fund research and they find that jointly these factors explain about 80% of hedge fund returns. Nevertheless, they also find that even after considering these risk factors hedge funds generate positive alphas for the full sample period. Kosowski *et al.* (2007) use bootstrapping and Bayesian approaches to address some of the limitations common in hedge fund research. They document significant alphas and also substantial persistence in alphas in hedge funds, which suggests that the superior performance of hedge funds cannot be solely attributed to luck. Similarly, Ibbotson *et al.* (2011) conclude that alphas earned by hedge funds are positive and remarkably stable over time even during a financial crisis.

In contrast, Malkiel & Saha (2005) and Getmansky *et al.* (2015) argue that after adjusting for database biases hedge funds on average underperform their benchmarks. Fung *et al.* (2008) observe a positive and statistical alpha only for an 18-month long subperiod out of the sample covering 120 months. Billio *et al.* (2014) conclude that the alphas generated by hedge funds change dramatically over time and across categories. Capocci & Hubner (2004) observe positive excess return for 10 out of 13 investment strategies that they analyze, but only for one quarter of individual hedge funds. They also show that best-performing funds follow momentum strategies and have limited holdings of emerging market bonds. Also Ding & Shawky (2007) suggest that the evaluation of hedge fund performance relative to market indices depends on the level of

aggregation of hedge fund data and on the adjustments for skewness in hedge fund returns distribution. They conclude that even though all hedge fund categories outperform the general market index less than half of the individual hedge funds beat it. Griffin & Xu (2009) find limited evidence of superior skills of hedge fund managers in timing the market and in picking individual stocks. The alphas they observe are small on a value-weighted basis and insignificant on an equal-weighted basis.

Some of the divergence in the reported results may be due to the data biases resulting from the voluntary nature of reporting of hedge fund performance in databases. A self-selection bias arises when successful hedge funds are more likely to report their performance to commercial databases. Jorion & Schwarz (2014) find that investment companies act strategically and they list in multiple commercial databases their small, best-performing funds, which helps them raise awareness about the funds and attract new investments (Fung & Hsieh, 1997, 2000). Agarwal *et al.* (2013) examine the impact of self-selection bias by comparing data in five commercial databases with information in Form 13F that are reported quarterly by advisors (rather than funds) with the Securities and Exchange Commission (SEC). They find that even though reporting initiation is more likely after a superior performance it subsequently declines. Similarly, Edelman *et al.* (2013) combine previously unexplored data sources with manual data collection to construct a comprehensive dataset of returns earned by large hedge fund management companies. Based on the sample covering more than half of the industry’s AUM they observe little differences between the reporting and non-reporting firms. In contrast, Aiken *et al.* (2013) use the mandatory regulatory filings by registered funds-of-funds (FoFs) and they observe that only about one-half of these fund-level returns are reported to one of the five major hedge funds databases.

The backfilling bias or the “instant-history bias” arises when hedge funds are included in databases together with their performance history only after succeeding during an “incubation period” intended to accumulate a performance track record before offering the fund to investors. Recording performance histories of only the successful funds introduces a positive bias into the database (Fung & Hsieh, 2000; Posthuma & Van der Sluis, 2003). To quantify its effect prior research compares returns generated in the first years of hedge fund existence in the database with other years. Estimates based on this approach range between 1.0% and 1.5% *per annum*

(Fung & Hsieh, 2000; Edwards & Caglayan, 2001).

The survivorship bias may arise when commercial databases terminate coverage of previously included funds. Providers may wish to purge the database of funds that no longer operate because they are not relevant to their clients anymore. Hodder *et al.* (2014) report that on average 15% of hedge funds exit the database every year. A bias arises when the funds that exit the database on average underperform the “surviving” funds. Brown *et al.* (1999) examine survivorship bias in a database of active and defunct offshore funds and observe positive risk-adjusted returns even after adjusting for the bias. Liang (2000) observes that poor performance is the main reason for a fund’s disappearance from the databases and finds that the survivorship bias exceeds 2% per annum and it varies with investment styles. Edwards & Caglayan (2001) compare the performance of defunct funds with those that are still in operation and they estimate the impact of the bias at 1.85% per annum. Agarwal *et al.* (2015) propose a range between 2.0% and 3.6% per annum.

The variability of prior empirical results and the potential impact of various data biases complicate the interpretation of this stream of research. Thus, we consider it worthwhile to conduct a quantitative survey to synthesize this pool of diverse empirical results and to examine how they are affected by publication selection and data biases. Our empirical approach builds on and complements earlier research that uses the meta-analysis methodology to study performance of other types of funds (Rathner, 2012; Revelli & Viviani, 2015; Coggin *et al.*, 1993). Rathner (2012) performs a meta-analysis of 500 performance estimates collected from 25 empirical studies on the performance of Socially Responsible Investment (SRI) funds and he concludes that most primary studies do not find any significant performance difference between the SRI funds and the conventional funds. Furthermore, similar to our paper, Rathner (2012) also studies the impact of treatment of the survivorship bias for the magnitude of the estimates reported in the primary studies and he concludes that primary studies that adjust for the survivorship bias are more likely to report SRI funds outperformance relative to conventional funds. Revelli & Viviani (2015) perform a meta-analysis of 85 studies and 190 experiments that examine the impact of SRI on financial performance. They conclude that, overall, SRI considerations have neither positive nor negative impact on performance. Coggin *et al.* (1993) conduct a meta-analysis of investment performance of U.S. equity pension funds and they examine their stock picking and

market timing abilities. They identify substantial differences in performance of individual funds and they observe that some funds produce substantial abnormal returns while others do not. They also conclude that regardless of the choice of benchmark portfolio or estimation model, equity pension funds exhibit superior stock-picking and inferior market-timing skills. Our paper builds on this prior research by applying these methodological approaches to measure hedge fund performance and it extends them by employing a battery of modern econometric techniques for identifying publication selection bias.

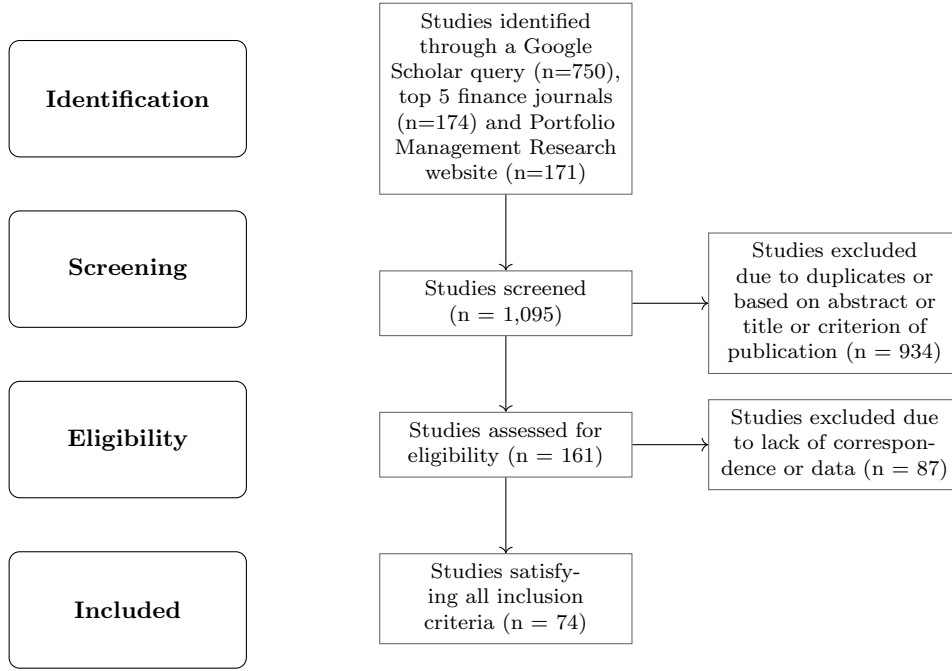
3 Dataset

To perform a comprehensive analysis of how published evidence on hedge fund performance is affected by selective publication and data biases we collect a large dataset of alpha estimates from primary studies. Alpha estimates represent abnormal returns adjusted for exposures to risk factors. Individual alpha coefficients reported in primary studies thus aim to capture the same underlying concept of value generated by hedge funds for investors. All the collected alpha coefficients are measured in the same unit (i.e. percentage) and they are normalized to monthly frequency. Hence, they are directly comparable, which makes them suitable for aggregation in a quantitative survey.

Our data collection process follows the guidelines proposed by Havranek *et al.* (2020). We restrict our analysis to estimates published in peer-reviewed research journals. The peer-review process constitutes an important quality assurance mechanism. Using only estimates that underwent the peer-review process increases the likelihood that the collected alpha coefficients are estimated using established methodological approaches and they are free of error. Furthermore, we expect most researchers and practitioners to form their subjective understanding of typical alpha estimates predominantly based on published articles. Our sample thus likely mirrors the set of studies that shape people’s views of hedge fund performance.

Figure 2 provides an overview of the individual steps of our data collection process. First, we built a preliminary list of studies based on references included in the sections on hedge fund performance in two comprehensive review articles: Connor & Woo (2004) and Agarwal *et al.* (2015). Second, we perform a systematic Google Scholar search using the following combinations of keywords: “hedge fund returns” OR “hedge fund performance”. We search for

Figure 2: PRISMA flow diagram



Note: We perform our primary search in Google Scholar based on the following combinations of keywords: “hedge fund returns” OR “hedge fund performance”. Furthermore, we perform our secondary search based on slightly broader set of keywords: “hedge fund” OR “hedge funds” in the following finance journals: the *Journal of Finance*, the *Journal of Financial Economics*, the *Review of Financial Studies*, the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*. Finally, we perform our tertiary search based on the combination of keywords: “hedge fund” OR “hedge funds” in the journals listed on the Portfolio Management Research website: the *Journal of Portfolio Management*, the *Journal of Financial Data Science*, the *Journal of Impact and ESG Investing*, and the *Journal of Fixed Income*. We screen for the alpha coefficients the first 750 studies identified by our primary Google Scholar search, as well as 174 studies identified by our secondary search in the top 5 finance journals, and additional 171 studies identified by our tertiary search in journals listed on the Portfolio Management Research website. We are left with 161 studies after the screening. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) is an evidence-based set of items for reporting in systematic reviews and meta-analyses. More details on PRISMA and reporting standard of meta-analysis in general are provided by Havranek *et al.* (2020).

alpha estimates in the articles as ordered by Google Scholar. We terminate this phase of data collection after having covered the first 750 articles in the Google Scholar list. We observe that after having reached this position at Google Scholar list the articles become less relevant and the likelihood of identifying additional articles with usable alpha estimates drops dramatically. Third, to make sure that our search does not miss any important articles we perform a slightly broader Google Scholar search using less restrictive keywords: “hedge fund” OR “hedge funds” in the following finance journals: the *Journal of Finance*, the *Journal of Financial Economics*,

the *Review of Financial Studies*, the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*. Fourth, to ensure a comprehensive coverage of articles in journals aimed primarily at investment professionals and which may not be as highly cited and ranked by Google Scholar we perform a similar search based on the combination of keywords: “hedge fund” OR “hedge funds” in the journals listed on the Portfolio Management Research website³: the *Journal of Portfolio Management*, the *Journal of Financial Data Science*, the *Journal of Impact and ESG Investing*, and the *Journal of Fixed Income*.

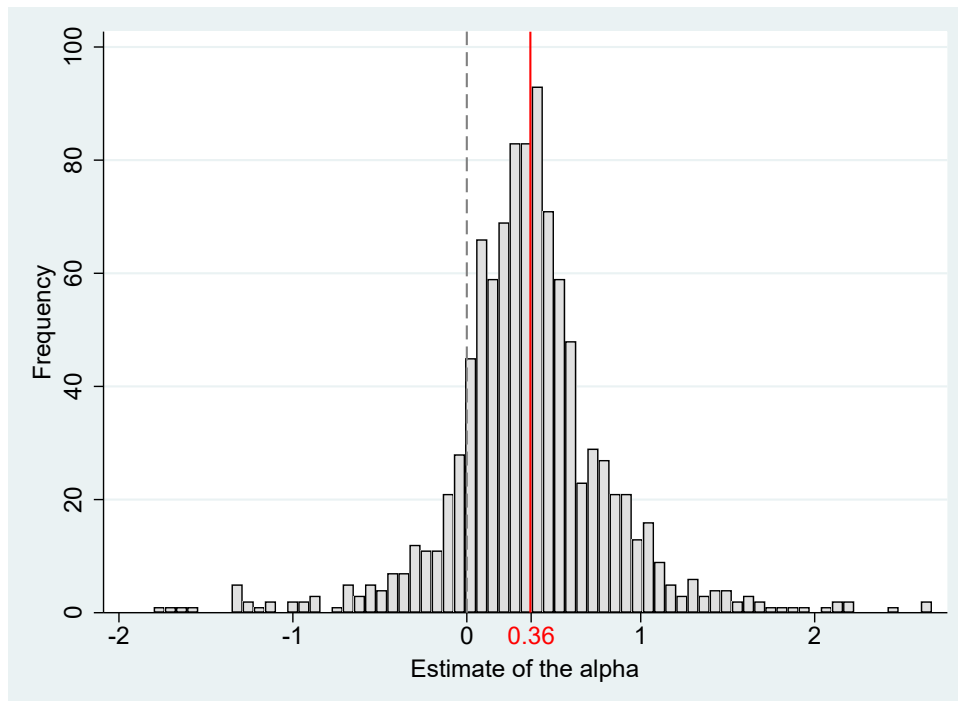
To be included in the dataset, a given alpha estimate must be accompanied by a measure of statistical significance, i.e. a t -statistic, a standard error (SE), and/or a p -value. We use these measures to compute the precision of individual alpha estimates. We use the precision variable in our tests of selective publication as well as for our data verification. Before constructing our final sample we attempt to identify alpha coefficients that may have resulted from human error in data hand-collection. To do so we first convert all the measures of statistical significance to a common metric, i.e. t -statistic. Whenever available we collect corresponding t -statistics from primary studies. If the authors report standard errors instead, we compute the implied t -statistic as the ratio of the alpha coefficient and the corresponding standard error. In studies using the Bayesian approach, we divide the alpha coefficient by the reported standard deviation. If the authors report p -values we check whether they explicitly state that these are based on a one-tailed or a two-tailed tests. If the type of the test is not explicitly stated in the article we try to infer it from the discussion of the level of statistical significance of results tabulated in the primary studies. If the type of the test cannot be ascertained from the interpretation of the results we assume a two-tailed test (1 study). We then use the inverse t -distribution to convert the reported p -value to a t -statistic. If the authors report the total number of observations based on which a given alpha coefficient is estimated we use that number for the degrees of freedom. If the authors report both the number of observations in the cross-section and in the time-series we use the product of the two numbers. If the information on the number of observations is only provided for the cross-section or the time-series, we use that number instead. If none of the above is provided we assume 168 observations, which is equal to the sample median for the subsample where the number of observations is explicitly stated. We then check all observations with the implied t -statistic greater than 10 for potential errors in hand-collecting the data. We

³Source: <https://www.pm-research.com/>).

ensure that such results are presented as highly significant in the main text of the primary study. We discard the 1 observation where the authors report a t -statistic greater than 50.

Our data collection procedure yields 1,019 alpha estimates obtained from 74 primary studies. The data sample size makes our study one of the largest quantitative surveys of prior studies in financial economics. The first alpha estimates that meet the sample collection criteria specified above were published in 2001. We end our data collection on September 1, 2021. The long time span exceeding 20 years ensures that our sample of alpha estimates is representative of the accumulated pool of evidence in this stream of research literature. Figure 3 shows the histogram of the alpha estimates in our sample. The figure suggests that the distribution is fairly normal and quite symmetric. Furthermore, we do not observe any significant kinks in the distribution, which indicates that no levels of alpha estimates are significantly underrepresented or over-represented. Figure 3 thus offers some preliminary indication that the distribution of our dataset has the expected characteristics and it is free from dramatic discontinuities.

Figure 3: Distribution of alphas



Note: The figure depicts a histogram of the alphas reported by individual studies. The solid red vertical line denotes sample mean. The gray dashed line denotes 0.

The vertical line in Figure 3 denotes the unconditional sample mean of monthly alphas of

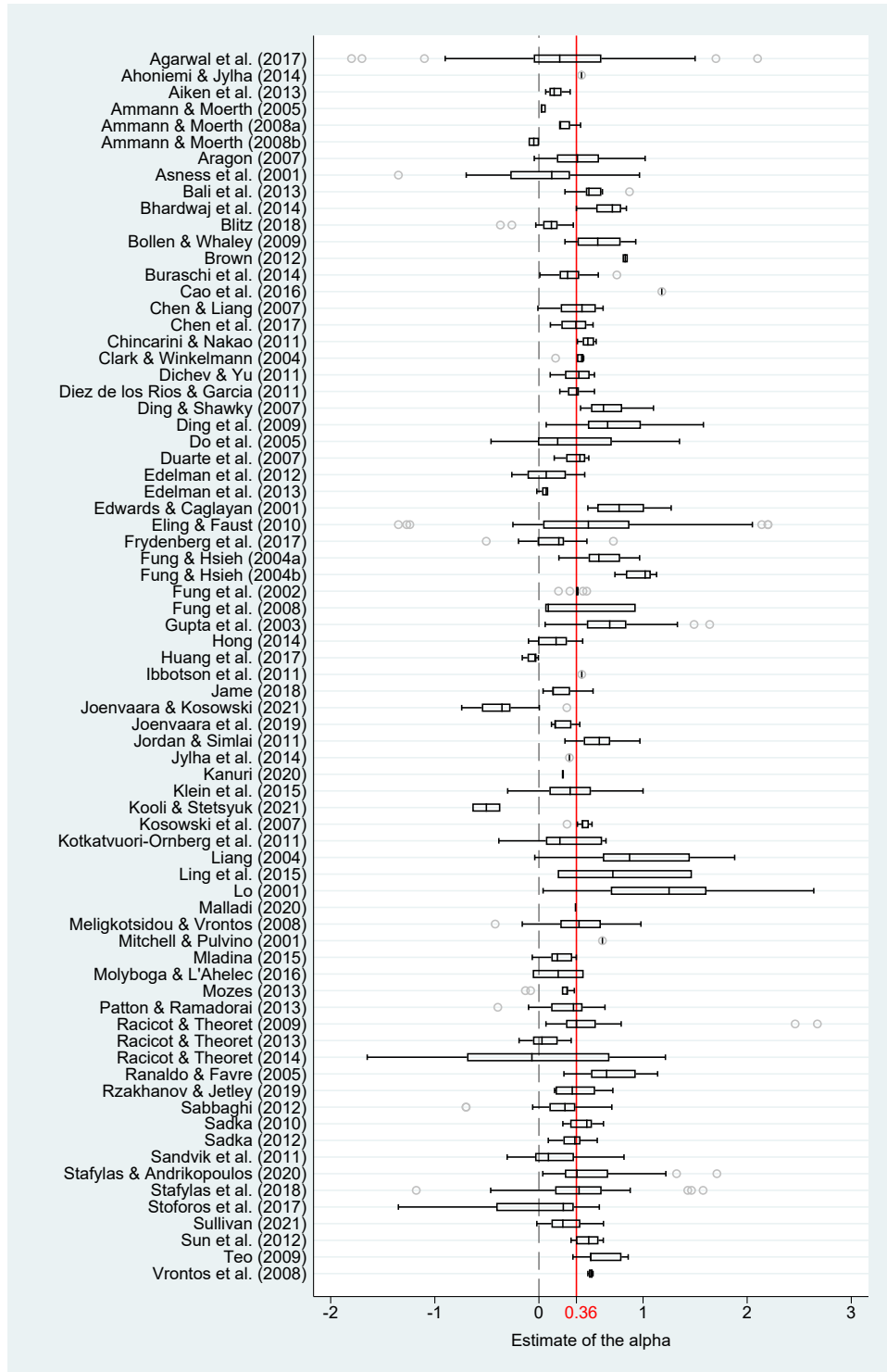
0.36%, which corresponds to annual abnormal return of 4.32%. This result is broadly consistent with values proposed in prominent studies on hedge fund performance. For example, Getmansky *et al.* (2015) report monthly alphas based on the Fung & Hsieh (2001) seven-factor model for various hedge fund strategies between 0.18% and 0.56%. This suggests that our dataset does not dramatically differ from what would be expected based on prior literature. At the same time the histogram shows that the individual alpha estimates are relatively dispersed. This suggests that there are substantial differences across various studies and estimation approaches.

Figure 4 visualizes the distribution of alpha estimates reported in the individual primary studies. The boxes represent the interquartile range between percentile 25 and percentile 75 and the vertical line inside each box denotes the median value for a given study. The whiskers represent the highest and lowest data points within 1.5 times the range between the upper and lower quartiles. Consistent with the preliminary indication in Figure 3, also the pattern depicted in Figure 4 shows that the individual studies differ greatly in the dispersion of reported alpha coefficients. For most studies, the interquartile ranges cross the vertical line representing the unconditional sample mean of 0.36%. However, there are some studies with interquartile ranges not overlapping with the unconditional sample mean. In fact, some of them are fully below zero. Furthermore, we observe that some studies exhibit rather wide interquartile ranges exceeding one percentage point. This suggests that even within some studies the reported coefficients vary greatly. The substantial heterogeneity of alpha estimates reported in primary studies further underscores the importance of conducting a quantitative survey that aggregates these diverse results and corrects them for potential biases.

We report further information about the characteristics of alpha coefficients in the individual primary studies in Table 1.⁴ The table illustrates substantial differences between the individual primary studies. The number of alphas collected from a study ranges from 1 to 61. The median primary study contributes nine alpha estimates to our sample. The number of databases the alpha estimates are based on also substantially varies across the individual primary studies. The median value of one indicates that a typical study uses only one database as a source of data. This observation again highlights the importance of aggregating and synthesizing the hedge fund performance estimates. Nevertheless, the most comprehensive studies include up to seven databases. The sample period of returns data in a typical study spans 171 months,

⁴We are grateful to an anonymous referee for proposing this overview table.

Figure 4: Alphas in primary studies



Note: The figure shows the distribution of alpha coefficients across the individual primary studies (sorted alphabetically). For each study, the box represents the interquartile range (P25-P75), and the dividing line inside the box shows the median value. The whiskers represent the highest and lowest data points within 1.5 times the range between the upper and lower quartiles. The solid red vertical line denotes the sample mean; the dashed gray vertical line denotes zero. For ease of exposition, outliers are excluded from the figure but included in all statistical tests.

which corresponds to more than 14 years. However some studies use data sets covering only 31 months (about 2.5 years), while others comprise 475 months, i.e., almost 40 years. Most studies use only one risk model for estimating the abnormal returns. However, some studies use up to seven risk models. We collect about 30% of our alpha estimates from studies published in the Top 5 finance journals (i.e., the *Journal of Finance*, the *Journal of Financial Economics*, the *Review of Financial Studies*, the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*).

Table 1 also reports the mean value, the standard deviation, the minimum, the median value, and the maximum of the alpha coefficients reported in a given primary study. The mean alpha estimate reported in a study ranges from -0.51 to 1.22. This suggests that the individual primary studies reach very different conclusions about the abnormal returns generated (or destroyed) by hedge funds. Our study aggregates these diverse findings and draws inferences about the overall value reported in this stream of research. The following section presents these aggregated results.

Table 1: Primary studies

Study	#Alphas	#Sources	#Months	#Models	Top5	Mean	StDev	Min	Md	Max
Agarwal <i>et al.</i> (2017)	61	1	240	3	1	0.27	0.77	-1.80	0.20	3.00
Ahoniemi & Jylha (2014)	1	1	180	1	0	0.41		0.41	0.41	0.41
Aiken <i>et al.</i> (2013)	6	6	72	3	1	0.22	0.12	0.09	0.20	0.40
Ammann & Moerth (2005)	3	1	136	2	0	0.03	0.02	0.02	0.03	0.06
Ammann & Moerth (2008a)	4	2	136	2	0	0.25	0.10	0.20	0.20	0.40
Ammann & Moerth (2008b)	2	1	136	2	0	0.00	0.01	-0.01	0.00	0.00
Aragon (2007)	16	1	96	2	1	0.40	0.30	-0.05	0.37	1.02
Asness <i>et al.</i> (2001)	20	1	81	2	0	0.00	0.51	-1.35	0.12	0.97
Bali <i>et al.</i> (2013)	11	1	216	1	0	0.51	0.16	0.25	0.48	0.87
Bhardwaj <i>et al.</i> (2014)	6	1	223	1	1	0.66	0.18	0.36	0.71	0.84
Blitz (2018)	19	2	204	1	0	0.09	0.17	-0.37	0.12	0.33
Bollen & Whaley (2009)	4	1	144	1	1	0.58	0.28	0.25	0.57	0.93
Brown (2012)	2	1	168	1	0	0.83	0.04	0.80	0.83	0.85
Buraschi <i>et al.</i> (2014)	45	1	198	3	1	0.30	0.16	0.01	0.28	0.75
Cao <i>et al.</i> (2016)	1	1	228	1	1	1.18		1.18	1.18	1.18
Chen & Liang (2007)	10	3	138	7	1	0.37	0.20	-0.01	0.41	0.62
Chen <i>et al.</i> (2017)	4	2	216	1	1	0.34	0.17	0.11	0.36	0.52
Chincarini & Nakao (2011)	7	1	475	3	0	0.47	0.07	0.37	0.47	0.55
Clark & Winkelmann (2004)	5	1	119	1	0	0.36	0.11	0.16	0.41	0.43
Dichev & Yu (2011)	8	2	348	2	1	0.36	0.15	0.11	0.38	0.53
Ding & Shawky (2007)	17	1	168	5	0	0.65	0.21	0.40	0.62	1.10
Ding <i>et al.</i> (2009)	9	1	144	1	0	0.73	0.45	0.07	0.66	1.58
Do <i>et al.</i> (2005)	26	1	31	1	0	0.34	0.51	-0.46	0.18	1.35
Duarte <i>et al.</i> (2007)	4	3	194	1	1	0.35	0.14	0.15	0.39	0.48
Edelman <i>et al.</i> (2012)	6	4	72	3	0	0.08	0.26	-0.26	0.07	0.44
Edelman <i>et al.</i> (2013)	6	4	108	1	1	0.06	0.04	-0.02	0.07	0.09
Edwards & Caglayan (2001)	9	1	104	1	0	0.82	0.27	0.47	0.77	1.27
Eling & Faust (2010)	30	1	152	6	0	0.47	0.91	-1.35	0.48	2.20
Frydenberg <i>et al.</i> (2017)	33	1	254	1	0	0.14	0.24	-0.51	0.19	0.72
Fung <i>et al.</i> (2002)	9	1	84	1	0	0.36	0.08	0.19	0.37	0.46
Fung & Hsieh (2004a)	22	4	108	1	0	0.59	0.23	0.19	0.58	0.97
Fung & Hsieh (2004b)	8	2	108	4	0	0.97	0.15	0.73	1.02	1.13
Fung <i>et al.</i> (2008)	3	3	57	1	1	0.36	0.49	0.06	0.09	0.93
Gupta <i>et al.</i> (2003)	42	1	152	2	0	0.73	0.35	0.06	0.68	1.64
Hong (2014)	6	1	65	3	0	0.15	0.19	-0.10	0.17	0.42
Huang <i>et al.</i> (2017)	5	1	174	1	0	-0.07	0.07	-0.16	-0.03	-0.01

(continued on next page)

Table 1: Primary studies (continued)

Study	#Alphas	#Sources	#Months	#Models	Top5	Mean	StDev	Min	Md	Max
Ibbotson <i>et al.</i> (2011)	1	1	180	1	0	0.41		0.41	0.41	0.41
Jame (2018)	10	2	212	2	0	0.20	0.14	0.04	0.14	0.52
Joenvaara <i>et al.</i> (2019b)	5	5	228	1	1	0.23	0.12	0.12	0.16	0.39
Joenvaara & Kosowski (2021)	17	7	120	5	1	-0.37	0.26	-0.74	-0.36	0.27
Jordan & Simlai (2011)	12	1	180	1	0	0.57	0.20	0.25	0.58	0.97
Jylha <i>et al.</i> (2014)	1	1	216	1	1	0.29		0.29	0.29	0.29
Kanuri (2020)	2	1	225	2	0	0.23	0.01	0.22	0.23	0.24
Klein <i>et al.</i> (2015)	30	4	108	1	0	0.34	0.34	-0.30	0.30	1.00
Kooli & Stetsyuk (2021)	2	1	289	2	0	-0.51	0.19	-0.64	-0.51	-0.37
Kosowski <i>et al.</i> (2007)	9	5	108	2	1	0.42	0.07	0.27	0.42	0.51
Kotkatvuori-Ornberg <i>et al.</i> (2011)	6	1	175	1	0	0.22	0.38	-0.39	0.20	0.64
Liang (2004)	20	1	36	5	0	1.11	0.77	-0.04	0.87	3.50
Ling <i>et al.</i> (2015)	3	1	67	2	0	0.79	0.65	0.18	0.71	1.47
Lo (2001)	11	1	47	1	0	1.22	0.70	0.04	1.25	2.64
Malladi (2020)	1	1	155	1	0	0.35		0.35	0.35	0.35
Meligkotsidou & Vrontos (2008)	20	2	143	1	0	0.37	0.34	-0.42	0.39	0.98
Mitchell & Pulvino (2001)	1	1	108	1	1	0.61		0.61	0.61	0.61
Mladina (2015)	10	1	240	1	0	0.19	0.14	-0.06	0.18	0.36
Molyboga & L'Ahelec (2016)	2	1	228	1	0	0.19	0.35	-0.06	0.19	0.43
Mozes (2013)	10	1	234	5	0	0.20	0.16	-0.13	0.27	0.34
Patton & Ramadorai (2013)	23	4	186	4	1	0.26	0.22	-0.39	0.33	0.63
Racicot & Théoret (2009)	47	1	108	1	0	0.57	0.83	0.07	0.36	5.11
Racicot & Théoret (2013)	5	1	183	1	0	0.05	0.20	-0.19	0.03	0.31
Racicot & Théoret (2014)	24	1	183	3	0	-0.07	0.85	-1.65	-0.07	1.22
Ranaldo & Favre (2005)	42	1	152	3	0	0.71	0.26	0.24	0.65	1.14
Diez De Los Rios & Garcia (2011)	11	1	99	1	0	0.35	0.09	0.20	0.36	0.53
Rzakhanov & Jetley (2019)	7	1	264	1	0	0.36	0.21	0.15	0.32	0.71
Sabbaghi (2012)	20	1	203	1	0	0.18	0.35	-0.70	0.25	0.70
Sadka (2010)	11	1	180	1	1	0.42	0.12	0.23	0.46	0.62
Sadka (2012)	12	1	192	1	0	0.33	0.13	0.09	0.35	0.56
Sandvik <i>et al.</i> (2011)	36	1	182	1	0	0.13	0.25	-0.31	0.09	0.82
Stafylas <i>et al.</i> (2018)	40	2	291	1	0	0.39	0.52	-1.18	0.39	1.58
Stafylas & Andrikopoulos (2020)	24	2	291	1	0	0.51	0.43	0.04	0.37	1.71
Stofofos <i>et al.</i> (2017)	26	1	237	1	0	-0.22	0.92	-3.64	0.21	0.58
Sullivan (2021)	28	2	312	5	0	0.26	0.18	-0.02	0.23	0.62
Sun <i>et al.</i> (2012)	10	1	192	1	1	0.47	0.12	0.31	0.48	0.62

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Table 1: Primary studies (continued)

Study	#Alphas	#Sources	#Months	#Models	Top5	Mean	StDev	Min	Md	Max
Teo (2009)	5	3	78	2	1	0.59	0.22	0.33	0.50	0.86
Vrontos <i>et al.</i> (2008)	5	1	144	5	0	0.50	0.02	0.47	0.51	0.52
Total #Alphas	1019									
Total #Studies	74									
Mean	13.77	1.70	168.99	1.95	0.30	0.37	0.28	-0.08	0.36	0.86
StDev	13.16	1.32	77.41	1.45	0.46	0.31	0.23	0.66	0.29	0.81
Min	1.00	1.00	31.00	1.00	0.00	-0.51	0.01	-3.64	-0.51	-0.37
Md	9.00	1.00	171.00	1.00	0.00	0.35	0.20	0.06	0.36	0.62
Max	61.00	7.00	475.00	7.00	1.00	1.22	0.92	1.18	1.25	5.11

Notes: The table shows descriptive statistics for the alpha coefficients reported in the individual primary studies (sorted alphabetically). *#Alphas* denotes the number of alpha estimates that meet our sample collection criteria reported in a given study. *#Sources* shows the number of databases the alpha estimates in the primary study are based on. *#Months* specifies the number of months of the data sample used in the primary study covers. *#Models* shows the number of risk models used to estimate the alphas in the primary study. *Top5* is the indicator variable equal to 1 if the study is published in the *Journal of Finance*, the *Journal of Financial Economics*, the *Review of Financial Studies*, the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*, and 0 otherwise. The *Mean*, *StDev*, *Min*, *Md*, and *Max* refer to the mean value, the standard deviation, the minimum, the median value, and the maximum of the alpha coefficients reported in the primary study.

4 Full Sample Results

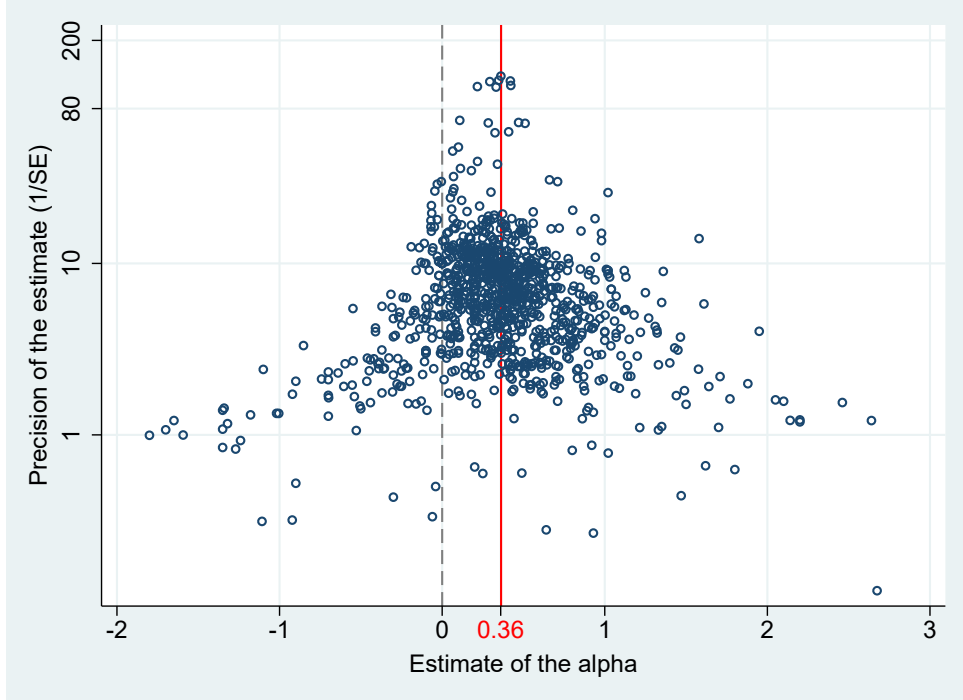
4.1 Funnel Plot

Having observed heterogeneity in reported alpha coefficients across various studies we now analyze whether these estimates are affected by publication selection bias. We start this analysis by visualizing how the alpha estimates reported in primary studies depend on their precision, which is defined as one over the estimate's standard error. Tests of statistical significance based on the t -distribution assume that the estimated coefficients and their standard errors are not correlated. Hence, in the absence of publication bias there should be no systematic relationship between the alpha coefficient and its standard error. In contrast, detecting a positive or a negative association between the coefficients and standard errors suggests selective publication. The authors of primary studies usually report t -statistics for their estimates, which implies that they assume that the estimates and their standard errors are statistically independent and the ratio of the estimates to their standard errors has a t -distribution. The association between the coefficient and its standard error can thus be used to detect selective publication.

In our setting the association can be depicted with a funnel plot with the alpha coefficients on the x-axis and their precision (i.e. $1/SE$) on the y-axis. We show such a funnel plot in Figure 5. In a bias-free world, the graph should resemble a symmetrical inverted funnel. The funnel shape arises because the most precise estimates tend to be concentrated around the underlying mean value, whereas less precise estimates with larger standard errors are more dispersed around the mean. The funnel plot shall be symmetric if for any given level of estimate precision both high and low estimates are equally likely to be published. In contrast, if imprecise estimates that are high tend to be reported, while equally imprecise estimates that are low get discarded then the funnel plot shall miss some observations in the left part and consequently it shall be asymmetric. An asymmetric funnel plot indicates that estimates are reported selectively in primary studies, which implies that their mean value provides a biased estimate of the underlying mean value in the population.

Figure 5 exhibits no obvious asymmetry, which is consistent with little or no publication bias. For any given level of precision both high and low estimates seem to be represented in the plot. The funnel plot thus provides initial suggestive evidence indicating that hedge fund alpha estimates reported in primary studies are not significantly contaminated by publication

Figure 5: Funnel plot of alphas



Note: When there is no publication bias, estimates should be symmetrically distributed around the mean denoted by the solid red vertical line. The gray dashed line denotes 0. Outliers are excluded from the figure for ease of exposition but included in all statistical tests.

selection bias. Furthermore, a simple visual examination of Figure 5 suggests that the funnel plot is slightly “hollow”, which might suggest that insignificant estimates (low precision alpha close to zero) are less likely to be published. Below we formally test for the significance of these observed patterns.

4.2 Formal Tests

Having provided preliminary evidence about the likelihood of a publication selection bias in hedge fund performance literature we proceed with using several approaches to formally test for it. The first set of tests exploits the above mentioned association between the alpha coefficients reported in primary studies and their standard errors. Since we use the term “alpha” to refer to the intercept term in the regression of returns on risk factors reported in primary studies, we use “kappa” to denote the constant (i.e. the intercept term) in our regressions of alpha coefficients on their standard errors. Furthermore, we use “lambda” to refer to the slope coefficient at the

explanatory variable of SE. We estimate the following equation:

$$\alpha_{ij} = \kappa + \lambda \cdot SE(\alpha_{ij}) + \epsilon_{ij}, \quad (2)$$

where α_{ij} stands for the i -th estimate of hedge fund alpha reported in the j -th study, $SE(\alpha_{ij})$ denotes its standard error, and ϵ_{ij} is the error term.

In the absence of any publication bias the slope coefficient λ is expected to be zero, which implies no association between the alpha estimates (α_{ij}) and their standard errors ($SE(\alpha_{ij})$). In contrast, if publication of estimated alpha coefficients is selective and low alpha estimates are more likely to remain unreported in primary studies then imprecise estimates (i.e. those with a large SE) should be more likely to be high rather than low leading to a positive λ coefficient. Conversely, a tendency to discard high rather than low alpha coefficients would lead to a negative λ coefficient. Hence, the slope coefficient λ reflects the effect of publication selection bias and the intercept term κ captures as the true mean alpha estimate corrected for the bias.

Panel A of Table 2 shows the results for several alternative ways of estimating Equation 2. In the first column we report the conventional ordinary least squares (OLS) estimate. As discussed above the OLS estimate represents the most straightforward way of testing for selective publication that is commonly used in prior research. However, it could yield spurious results in case unobserved features of the primary study design are correlated with the reported alphas. To address this potential problem we complement the OLS estimates with several alternative estimation techniques. The results reported in the second column of Table 2 are based on an estimation that includes study-level fixed effects (FE). Including study-level fixed effects filters out idiosyncratic study-level variation. Hence, as long as alpha estimates in a given primary study are estimated using similar methodologies including study-level fixed effects removes potential confounding effect of these methodological choices on the reported alpha estimates. Identification of the fixed-effect estimator rests on studies that report more than one estimate. Thus we complement the analysis with study-level between-effect estimation (BE) that accounts for the differences in study size. We report these results in the third column of Table 2.

To further address the issue of potential endogeneity in the method choices and reported standard errors in the primary studies we follow Stanley (2005) and Havranek (2015) and we

Table 2: Full sample results

<i>Panel A: Linear models</i>						
	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.0152 (0.188) [-0.534, 0.455]	-0.0265 (0.215)	0.0602 (0.131)	0.178 (0.353) [-0.526, 0.971] {-0.626, 0.983}	0.324 (0.320) [-0.415, 1.120]	0.0497 (0.127) [-0.348, 0.457]
Effect beyond bias (κ)	0.366*** (0.0426) [0.277, 0.458]	0.369*** (0.0540)	0.350*** (0.0474)	0.316*** (0.0854) [0.157, 0.475]	0.301*** (0.0440) [0.186, 0.412]	0.353*** (0.0380) [0.270, 0.436]
First-stage robust F-stat				12.71		
Studies	74	74	74	73	74	74
Observations	1,019	1,019	1,019	979	1,019	1,019
<i>Panel B: Non-linear models</i>						
	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.183* (0.106)	P = 0.631 (0.092)	L = 0.364 (p = 0.834)
Effect beyond bias	0.310*** (0.026)	0.325*** (0.009)	0.355*** (0.093)	0.320*** (0.008)	0.274*** (0.03)	0.386*** (0.045)
Studies	74	74	74	74	74	74
Observations	1,019	1,019	1,019	1,019	1,019	1,019

Notes: The first two panels report the results of the regression $\alpha_{ij} = \kappa + \lambda \cdot SE(\alpha_{ij}) + \epsilon_{ij}$, where α_{ij} denotes the i -th alpha coefficient estimated in the j -th study, and $SE(\alpha_{ij})$ denotes its standard error. FE: study-level fixed effects, BE: study-level between effects, IV: the inverse of the square root of the number of observations is used as an instrument for the standard error, WLS: model is weighted by the inverse of the standard error of an estimate, wNOBS: model is weighted by the inverse of the number of estimates per study. In Panel B, Top10 is model by Stanley *et al.* (2010), WAAP stands for Weighted Average of the Adequately Powered model by Ioannidis *et al.* (2017), Kinked-meta is endogenous kink model by Bom & Rachinger (2019), Stem model is by Furukawa (2020), selection is model by Andrews & Kasy (2019) using clustered SEs, P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published (normalized at 1), p-uni* is by van Aert & van Assen (2020), L denotes test statistic of p-uniform's publication bias test. Standard errors, clustered at the study level, are reported in parentheses. 95% confidence intervals from wild bootstrap in square brackets (Roodman *et al.*, 2018). In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

use the inverse of the square root of the number of observations in primary studies as an IV for the standard error. This measure has the desirable characteristics for a valid instrument. By construction, the number of observations is correlated with the standard error. At the same time, it is plausibly unrelated to the chosen estimation technique. Furthermore, it seems reasonable to assume that the number of observations is quasi-randomly distributed among the primary studies. The results based on this instrument thus constitute an important robustness check. We report these results in the fourth column in Table 2.

In the last two columns of Panel A of Table 2 we report our weighted least squares estimates of Equation 2. In the fifth column we weigh the observations by the inverse of their standard error (WLS). This approach gives less weight to less precise estimates, which helps adjusting

for potential heteroskedasticity in our observations. The sixth column shows our results from estimation when the observations are weighted by the inverse of the number of estimates reported in a given study (wNOBS). This approach provides a more comparable basis for larger and smaller studies.

Considering the above-discussed results reported in Panel A of Table 2 we find little evidence of publication selection bias for the alpha estimates reported in the primary studies. The λ coefficients that capture the effect of selective publication are all small in magnitude and statistically insignificant. These findings are remarkably consistent across the alternative ways of estimating Equation 2. Thus, consistent with our preliminary findings based on the funnel plot in Figure 5, our formal tests provide evidence consistent with non-selective publication of estimated monthly alphas in the primary studies in our sample.

Panel A of Table 2 also shows the κ estimates that reflect the estimated magnitude of the monthly alphas adjusted for the publication selection bias. We observe that these estimates range between 0.301 and 0.369 and they are strongly statistically significant at a better than 1 percent level in all specifications. It is noteworthy that the unconditional mean of monthly alpha estimates of 0.36 highlighted in the histogram in Figure 3 falls within this range of estimates corrected for the publication selection bias. Considering perhaps the most conservative estimate reported in Panel A of Table 2 we observe that the lower bound of the bootstrapped confidence interval of the IV specification is 0.157. This suggests that the true hedge fund alpha is unlikely to be below 1.9% per annum (0.157×12). These results further strengthen our earlier conclusion that the alpha estimates in our sample are not contaminated by selective publication and hedge funds do earn positive alphas for their investors.

Our estimates of Equation 2 reported in Panel A of Table 2 are subject to several limitations. First, these tests of selective publication are based on an assumption of a linear relationship between the estimate and its standard error. In reality, this association may not be linear. For example, it may exhibit discontinuities around conventional levels of statistical significance, i.e. when the t -statistics approaches 1.96. Second, the IV specification may not fully remedy the endogeneity problem because it may arise for reasons other than the bias due to omitted variable related to the research design in primary studies. Gechert *et al.* (2022) point out that endogeneity may arise even when deliberately reporting spuriously precise estimates, for example

due to reverse causality. Furthermore, since the standard error is itself an estimate endogeneity can also manifest itself through the measurement error. We address these shortcomings in Panel B of Table 2. To address the first issue, we use nonlinear techniques for publication bias correction. To address the second limitation, we use the p-uniform* approach recently developed by van Aert & van Assen (2020) that does not rely on the assumption of exogeneity.

In the first column of Panel B of Table 2 we report results based on the Top10 method proposed by Stanley *et al.* (2010). The method is based on a simple proposition that the bias arising from aggregating potentially selectively reported coefficients can be addressed by simply considering only the 10% most precise estimates. The second column of Panel B of Table 2 shows results based on the Weighted Average of the Adequately Powered (WAAP) model proposed by Ioannidis *et al.* (2017). Similarly to Top10 also WAAP is based on averaging only a subset of published coefficients. Ioannidis *et al.* (2017) examine statistical power of the results published in the field of economics and they propose dropping all estimates with statistical power lower than 80% and weighting the remaining estimates by the inverse of their variance. In the third column of Panel B of Table 2 we report the results from the Stem-based method recently developed by Furukawa (2020). The stem-based method builds on Stanley *et al.* (2010) but it aims at limiting the loss of sample variation that results from discarding 90% of the less precise estimates. Furukawa (2020) optimizes the trade-off between the bias and variance, discards only estimates that do not add value in the light of this trade-off and uses the remaining estimates to compute the average value. The first three columns in Panel B of Table 2 show estimates ranging from 0.310 to 0.355, which falls within the range documented earlier for the linear methods reported in Panel A (0.301 to 0.369). Thus, even based on these alternative methods we reach a similar conclusion on the limited impact of publication bias and on the values of alpha estimates corrected for a potential publication bias.

In the fourth column of Panel B of Table 2 we report results based on the endogenous kink model (Kinked-meta) proposed by Bom & Rachinger (2019). The model is based on the assumption that the relationship between an estimate and its standard error is only linear to some point because for some levels of reported coefficients there is no reason to expect the presence of publication bias. Hence, there is an endogenously determined cut-off value (or a “kink”) at which the relationship changes. The Kinked-meta model yields some weak

evidence (significant at 10% level) on the presence of selective publication (λ 0.183, SE 0.106). Nevertheless, even this approach yields a κ coefficient of 0.320, which is very close to the uncorrected mean of 0.36 and comfortably within the interval of 0.301 and 0.369 shown in Panel A.

The fifth column of Panel B of Table 2 shows our results for the selection model recently developed by Andrews & Kasy (2019). The model is based on the assumption that the probability of publishing an estimate depends on its statistical significance. The model identifies how likely it is for an estimate to fall into different intervals determined by the critical values of t -statistics. The model gives more weight to intervals that are underrepresented. Our results from the selection model suggest that statistically insignificant estimates may be somewhat less likely to get published than statistically significant estimates (63% probability vs. 100% probability). However, the corrected mean of alpha estimates decreases only slightly to 0.274. Hence, even this methodological approach does not suggest that inferences about the magnitude of alpha coefficients are greatly affected by selective publication.

The Andrews & Kasy (2019) model relies on several assumptions. It requires the estimates and their standard errors are statistically independent. It also assumes that the probability of publication is the same for all estimates in a given interval. We test these assumptions in Table A1 using the Kranz & Putz (2022) framework. These tests suggest some of the underlying assumptions of the selection model (especially the independence assumption) may be violated in many of our samples. Therefore, as robustness checks, we also use models that do not rely on the underlying assumption of no correlation between the estimates and their standard errors in the absence of publication bias.

The last column in Panel B of Table 2 shows the results of the p-uniform* model by van Aert & van Assen (2020). Harvey *et al.* (2016) and Harvey (2017) suggest that “ p -hacking”, i.e. a greater tendency to publish statistically significant rather than insignificant results, is a major problem in financial economics research. They argue that “ p -hacking” may have led to a number of “false positives” to be reported in published research on factors explaining the cross-sectional variation in realized returns. This introduces important distortions in modeling systematic risk in asset pricing, because the association between many of these variables and realized returns is unlikely to persist in the future. Following this argument, we examine potential “ p -hacking” in

research on hedge fund performance. To do so, we use the p-uniform* model recently proposed by van Aert & van Assen (2020) that is based on evaluating the distribution of p-values around the 5% cut-off level that is conventionally used to assess statistical significance. A tendency to publish statistically significant results implies over-representation of p-values just below the 5% cut-off and under-representation of p-values just above it. P-uniform* corrects for this potential bias by assigning different weights to estimates of various degrees of statistical significance based on the estimated publication probability. This selection model is robust to the assumption of zero correlation between estimates and standard errors in the absence of any publication bias. Our results based on this methodological approach are consistent with our previous findings. The test statistic for the publication bias (denoted “L”) is statistically insignificant, which again suggests that publication of alpha estimates in primary studies is not selective. In fact, this method suggests a somewhat higher value of 0.386 for alpha estimates corrected for the publication bias. Thus, we reach similar conclusions about the absence of publication selection bias and a somewhat similar estimate of the true mean value of the alpha coefficient even when using the p-uniform* method, which does not require the exogeneity assumption for the standard errors to be satisfied.

In contrast to the more conventional linear approaches reported in Panel A of Table 2, the more sophisticated non-linear approaches shown in Panel B of Table 2 do not require the linearity and exogeneity assumptions to be met. Overall, these approaches lead to fairly similar conclusions about the limited impact of selective publication on the alpha estimates reported in primary studies. Only the Kinked-meta model shows some marginally significant evidence of publication selection bias. However, even this approach does not dramatically alter the estimated value of alpha coefficients corrected for the publication bias. Furthermore, the selection model suggests that statistically insignificant estimates may be somewhat less likely to get published. However, the estimate for the mean alpha coefficient does not dramatically change after correcting for this bias. The interval of corrected alpha estimates based on the more sophisticated approaches reported in Panel B is slightly wider and it ranges from 0.274 to 0.386. However, both the upper bound and the lower bound of this interval are fairly close to the unconditional mean of 0.36. These results thus provide further support for our conclusion that inferences about the magnitude of the alpha coefficient in the literature on hedge fund

performance are not significantly affected by publication selection bias.

To complement our analysis we examine whether our results may be affected by the inclusion of studies that are coauthored by similar research teams. We acknowledge that different teams of co-authors may plausibly have various preferences over the choice of estimation methodology and they may have access to different data sources. Hence, the alpha coefficients estimated by members of a given research team may be interdependent.⁵ To address this issue, we recompute our results and we cluster the standard errors by author teams. We report these results in Table A2. These results show that potential inter-dependencies between the alphas estimated in individual research teams do not materially affect our conclusions. Consistent with our main results, we find little evidence of publication selection bias after clustering the standard errors at the research team level. All the λ coefficients reported in Table A2 are statistically insignificant, which suggests against selective publication. Furthermore, the κ coefficients fall within a fairly narrow interval of (0.301, 0.369) that is fully subsumed by the corresponding interval that we observe for our main results (0.274, 0.386). We thus conclude that clustering of standard errors at the level of a research team leads us to similar conclusions on the absence of selective publication and on the magnitude of the “corrected” alpha estimate as our main results.

Finally, we follow recent advances in econometrics and conduct a test of “*p*-hacking” based on Elliott *et al.* (2022). We report the results in Figure A1 and Table A3. The “*p*-hacking” tests are conceptually different from the publication bias tests. Therefore, they constitute a good complement to the results reported earlier in this section. Figure A1 shows no obvious breaks at the value of 1.96 (represented by the vertical solid red line), which represents the most important threshold for statistical significance at 5% level. Similarly, our formal test show no indication of over-reporting of statistically significant estimates in primary studies (see Table A3). Consistent with our earlier conclusions, these results also suggest that the pool of empirical evidence on hedge fund performance is not substantially contaminated by selective reporting of estimates in research journals.

We find these results remarkable, especially when contrasted with the abundant empirical evidence on the prevalence of publication selection bias in a multitude of other settings in economics and finance, e.g. Stanley (2001, 2005); Stanley & Doucouliagos (2010); Havranek (2015);

⁵We are grateful to an anonymous referee for pointing this out.

Brodeur *et al.* (2016); Bruns & Ioannidis (2016); Stanley & Doucouliagos (2017); Christensen & Miguel (2018); Brodeur *et al.* (2020); Blanco-Perez & Brodeur (2020), and Zigraiova *et al.* (2021). We can only speculate about the underlying reasons why we do not observe selective publication in research on hedge fund performance. We consider it likely that the presence of two opposing perspectives, both of which may be quite plausible, limits researchers' and editors' incentives to systemically discard either high or low estimates of hedge fund performance. On the one hand, hedge funds likely employ very talented wealth managers who are highly incentivized to generate returns for investors. Hence, it may be reasonable to expect that these bright and highly motivated minds are capable of identifying assets that are temporarily mispriced due to investor irrationality or the impact of passive investment. Hedge funds can possibly earn abnormal returns by investing in these assets, which implies positive alpha coefficients. On the one hand, following the EMH, hedge funds mostly trade on competitive markets where it may be challenging to systematically earn more than the "normal" rate of return. In addition, the high management and performance fees that hedge funds charge may imply that their net-of-fees performance may be inferior to passive indexing. This would imply either insignificant or negative alphas. We consider it plausible that the lack of a clear *a priori* theoretical prediction about the expected sign of estimated coefficients may limit the incentives for selective reporting and increase the readiness of academic journals to publish both positive and insignificant or negative results on hedge fund performance.

To further strengthen our analysis, we perform several robustness checks intended to ensure that our results are not driven by the heterogeneity in the mix of various alpha coefficients estimated in the primary studies using a wide range of techniques. Heterogeneity in estimation may potentially lead to offsetting biases that would compromise our ability to detect selective publication in the full sample. For example, it is acknowledged that the p-uniform* method tends to overestimate the measured effect when large heterogeneity is present among the estimates collected from the primary studies (Carter *et al.*, 2019). To further strengthen the confidence in our findings and to rule out the possibility that our tests are adversely affected by the diversity of the techniques used in estimating the alpha coefficients in the primary studies we proceed by analyzing more homogeneous subsets of alpha estimates to determine whether selective publication can be observed in any of these subsamples.

5 Subsample Results

In this section, we report our results for various subsamples of our dataset. Our data contain alphas estimated using different data sources and estimation techniques. We argue that it is important to aggregate these different estimates and report the representative alphas, as all estimates reported in primary studies are likely to help shape researchers' and investors' views of the abnormal returns that hedge funds earn on average. We assume that different estimation approaches are used in the literature to date because opinions differ about their relative appropriateness. We assume that readers of the research literature on hedge fund performance have differing views on these techniques and rely most heavily on the alphas estimated using the methods they believe to be the best. Our analysis of the full sample recognizes that none of the approaches is universally superior and takes into account all of the estimates that are likely to shape researchers' and practitioners' views on the subject. Despite this advantage, the overall results may be affected by the diversity of the underlying data and their impact on the power of our test. To ensure that our results are not affected by the underlying heterogeneity of the data, we re-estimate our regressions for more narrowly defined and thus more homogeneous subsamples. We observe whether the results of our subsample are consistent with the results of the whole sample.

We consider several subsamples of more homogenous alpha estimates. First, we partition our sample based on whether the survivorship and/or backfilling biases are adjusted for in a given primary study. Since these data biases may potentially have a significant impact on the documented returns estimating our regressions for the two subsamples separately lets us draw stronger inferences from our results. Second, we consider alpha coefficients estimated using two commonly used risk models: (i) the one-factor model, and (ii) the seven-factor model. Hedge funds exhibit unusual risk exposures to various risk dimensions and so the choice of a risk model may have an impact on the estimated abnormal return. Third, we recompute our results for the subsample of alpha coefficients estimated with the use of IV. Prior research shows that IV-based estimates are more likely to suffer from a publication bias because they tend to have larger standard errors (Brodeur *et al.*, 2020). We examine whether we detect selective publication in this subset of estimates. Fourth, we report our results for the subsample of alpha estimates published in the Top 3 (the *Journal of Finance*, the *Journal of Financial Economics*, the *Review*

of *Financial Studies*), or the Top 5 (plus the *Journal of Financial and Quantitative Analysis*, and the *Review of Finance*) finance journals. These journals are particularly prominent in the field of finance, and so the alpha estimates they publish are likely to be particularly impactful in shaping researchers' views of hedge funds performance. In addition, many researchers are strongly incentivized to publish their research by these leading journals. This implies that if publication of hedge fund performance is selective, the bias is likely to be particularly strong in these journal.

To pave the way for computing our subsample results, we visualize in the Appendix the distribution of the alpha coefficients in these subsamples. Figure A2 shows the histograms and Figure A3 shows the funnel plots for the individual subsamples we analyze in this section. In the histograms, we do not observe any major deviations from normality. In a similar vein, with the exception of alpha estimates estimated with the use of IV-based methods, the funnel plots are rather symmetric, which points towards little publication selection bias. In the following subsections, we formally test for selective publication in these subsamples.

5.1 Survivorship and Backfilling Biases

Prior research has long argued that the survivorship and backfilling biases may have a substantial impact on hedge fund performance estimates (Fung & Hsieh, 2004a; Aggarwal & Jorion, 2010; Kosowski *et al.*, 2007). The survivorship bias arises when a data sample excludes performance results of funds that are no longer in existence. From the perspective of data providers, excluding these funds from their database is sensible because funds that no longer operate are not interesting for investors any more. Nevertheless, since performance of funds that stop reporting information on their performance to the database may systematically differ from performance of surviving funds purging this information biases the research results based on the database. The backfilling bias arises when funds undergo an “incubation period” intended to accumulate performance track record before they are offered to investors. If performance history is backfilled into the database only for those funds that succeed in the incubation period the database overstates performance of the entire hedge fund population in the early years of their existence.

In the following analysis we consider separately a subsample of alpha estimates that explicitly

controls for the survivorship and/or backfilling biases. Then we consider only those alpha estimates that do not adjust for these biases. Given that the survivorship and backfilling biases may have a significant impact on the estimated alpha coefficients considering only one subsample at a time makes the individual alpha estimates more homogenous. We examine whether our main conclusions on the limited publication selection bias are robust to testing these relationships within the two subsamples.

In Table 3, we report our results for two subsets of alpha estimates: (i) those adjusted for the survivorship and/or backfilling bias are reported in Part 1, and (ii) those adjusted for neither survival bias nor backfilling bias are reported in Part 2. The results based on the conventional approaches reported in Panel A of Part 1 convey a fairly similar albeit slightly weaker message as the results based on the full sample (reported in Table 2). In Panel A of Part 1 all but one λ coefficient are statistically insignificant. Only for the WLS model, which weighs the observations by the inverse of their standard error, we observe a significant λ coefficient of 0.472 (SE 0.233, significant at 5% level). This finding provides limited evidence on some publication bias within the subsets of alpha estimates adjusted for the survivorship and/or backfilling biases weighted by their precision. Furthermore, among the results based on the non-linear approaches reported in Panel B of Part 1 we observe one statistically significant λ for the Kinked-meta model of 0.519 (SE 0.234, significant at 1% level). Similarly, to the WLS model also Kinked-meta attributes different weights to alpha estimates based on their precision. However, overall, we find only limited evidence of selective publication of alpha coefficients within the subsample of estimates that adjust for the survivorship and/or backfilling biases.

Results presented in Part 1 of Table 3 also indicate that limiting the analysis to the subsample of estimates adjusted for the survivorship and/or backfilling biases does not dramatically affect the conclusions about the magnitude of the alpha coefficients. The κ coefficients reported in Part 1 reflect the average alpha coefficients adjusted for the publication selection bias range between 0.194 and 0.351. This range is only slightly lower than the corresponding interval for κ coefficients based on the full sample between 0.274 and 0.386 reported in Table 2. The most significant deviation from this pattern is the slightly lower and only marginally significant κ coefficient based on the IV estimate that uses the inverse of the square root of the number of observations as an instrument for the standard error. This κ of 0.194 (SE 0.106, significant at

Table 3: Survivorship and backfiling biases

Part 1: Survivorship and/or Backfiling Bias Treated						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.00148 (0.180) [-0.435, 0.568]	-0.0846 (0.0961)	0.0795 (0.189)	0.493 (0.402) [-0.492, 2.534] {-0.342, 1.566}	0.472** (0.233) [-0.092, 0.948]	0.0814 (0.195) [-0.406, 0.561]
Effect beyond bias (κ)	0.329*** (0.0373) [0.243, 0.411]	0.351*** (0.0252)	0.300*** (0.0622)	0.194* (0.106) [-0.104, 0.392]	0.241*** (0.0337) [0.133, 0.377]	0.300*** (0.0487) [0.183, 0.403]
First-stage robust F-stat				11.29		
Studies	50	50	50	49	50	50
Observations	605	605	605	565	605	605
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.519*** (0.125)	P = 0.632 (0.115)	NA (NA)
Effect beyond bias	0.267*** (0.028)	0.248*** (0.017)	0.220*** (0.068)	0.234*** (0.012)	0.262*** (0.029)	0.325*** (0.048)
Studies	50	50	50	50	50	50
Observations	605	605	605	605	605	605
Part 2: Survivorship and Backfiling Biases Untreated						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.0168 (0.363) [-0.941, 0.870]	0.0491 (0.486)	0.0602 (0.170)	-0.466 (0.725) [NA] {-5.850, NA}	0.438 (0.492) [-0.558, 1.749]	0.0337 (0.140) [-0.844, 0.671]
Effect beyond bias (κ)	0.416*** (0.0782) [0.245, 0.615]	0.400*** (0.114)	0.436*** (0.0686)	0.521*** (0.163) [NA]	0.334*** (0.0559) [0.112, 0.482]	0.443*** (0.0563) [0.326, 0.554]
First-stage robust F-stat				3.76		
Studies	29	29	29	29	29	29
Observations	414	414	414	414	414	414
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.343* (0.191)	P = 0.719 (0.124)	NA (NA)
Effect beyond bias	0.301*** (0.038)	0.359*** (0.012)	0.331*** (0.040)	0.351*** (0.009)	0.282*** (0.086)	0.507*** (0.077)
Studies	29	29	29	29	29	29
Observations	414	414	414	414	414	414

Notes: Part 1: Sample in which both biases are treated for (either the survivorship or the backfiling bias is treated or both biases are treated for). Part 2: Sample in which biases are not treated for (neither the survivorship nor the backfiling bias is treated for). NA = non-convergence to the result. [NA] = confidence interval could not be bounded. $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

10% level) is reported in the fourth column in Panel A of Part 1.

In Part 2 of Table 3 we report results based on the subsample of alpha estimates that do not explicitly control neither for the survivorship nor for the backfiling biases. The conclusion based on this subsample are very similar to the main results reported in Table 2. In line with

the full sample results, the λ coefficients are statistically insignificant with the exception of the one based on the Kinked-meta model, which is equal to 0.343 and similarly to the full-sample result it is marginally significant at 10% (SE 0.191). Furthermore, the κ coefficients reported in Part 2 of Table 3 range between 0.282 and 0.521. Relative to the corresponding range for the κ coefficients based on the full sample this range is slightly wider. The difference is mainly driven by the higher upper bound, which is consistent with the proposition that studies that control for the survivorship and/or backfilling biases tend to report lower alpha estimates than those that do not. Overall, these findings suggest that the alpha coefficients that are not adjusted for backfilling and survivorship biases are not reported selectively.

5.2 Risk Models

One of the key methodological issues in hedge fund performance research concerns the choice of risk models used to adjust for the systematic risk that a given investment strategy involves. These models define the risk dimensions considered relevant for a given investment strategy. Prior hedge fund performance research uses several risk models. Models that feature fewer risk factors (e.g. the CAPM, the three-factor, and the four-factor model) are well-established in general asset pricing and investment research, which implies that the alpha coefficients based on these models are easily comparable with alpha coefficients estimated to evaluate performance of other types of investments, e.g. mutual funds. On the other hand, hedge funds commonly employ complex investment strategies that may exhibit unusual risk profiles and exposures to risk dimensions that are not essential for conventional asset classes. Thus, standard risk models may not fully capture the exposure of hedge funds' investment strategies to all relevant risk dimensions. More complex risk models featuring additional risk dimensions designed specifically to measure hedge fund performance may thus be more effective in capturing the plurality of risk exposures that hedge fund strategies involve. The choice of a risk model is thus one of the important drivers for the heterogeneity in the alpha coefficients that we collect from primary studies. In evaluating the robustness of our findings to various ways of reducing heterogeneity in our sample we re-estimate our regressions using two subsamples of alpha coefficients estimated based on two frequently used risk models: (i) the one-factor model, and (ii) the seven-factor model.

Part 1 of Table 4 shows the results of our tests of selective publication for the subset of alpha coefficients based on the one-factor model. In these tests we include all the alpha estimates that use a single risk factor based on market portfolio returns, i.e. both the estimates that use raw market returns and those that use market returns in excess of the risk-free rate. These methodological modifications are relatively small and so we do not expect them to have a substantial impact on the reported alpha coefficients. In line with our earlier results, we do not find evidence of a significant publication bias for these narrowly defined subsample of alpha coefficients. The λ coefficients that capture the impact of a potential publication bias are all statistically insignificant. This suggests that the alpha estimates based on a one-factor model are not reported selectively in prior literature.

Part 1 of Table 4 also shows that the κ coefficients that reflect the estimated true magnitude of the alpha estimates corrected for the potential bias range from 0.349 to 0.707. This interval includes the unconditional mean of all the alpha estimates in our sample of 0.36. However, these estimates are somewhat higher than the ones we document for the full sample in Table 2. This may suggest that the single market-based risk factor does not fully control for the systematic risk hedge fund strategies involve and so the abnormal return based on the model is higher. Overall, these results provide additional support for the conclusion that the alpha estimates reported in prior literature are not subject to selective publication.

Prior literature acknowledges that the complexity and the dynamic nature of hedge funds' investment strategies may induce exposure to risk dimensions that are not included in conventional risk models. Prior research thus proposes alternative risk models designed specifically for investment strategies common in hedge funds. The most notable example of these models is the seven-factor model (Fung & Hsieh, 2004a; Fung *et al.*, 2008). The model comprises the following risk factors: (i) the stock market excess return, (ii) the spread between the small capitalization and large capitalization stock returns, the excess return pairs of look-back call and put options (iii) on currency futures, (iv) on commodity futures, and (v) on bond futures, (vi) the duration-adjusted change in the yield spread of the U.S. 10-year Treasury bond over the 3-month T-bill, and (vii) the duration-adjusted change in the credit spread of Moody's BAA bond over the 10-year Treasury bond. These risk factors are intended to capture risk exposures of a broad set of hedge fund types ranging from equity long-short funds to managed futures

Table 4: Risk models

Part 1: One-Factor Model						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.456 (0.488) [-1.649, 1.468]	-0.328 (0.657)	0.0326 (0.494)	-1.115 (0.819) [NA] {-3.629, 0.102}	0.453 (0.602) [-0.776, 2.269]	-0.338 (0.249) [-1.247, 0.909]
Effect beyond bias (κ)	0.562*** (0.0447) [0.465, 0.642]	0.534*** (0.145)	0.411*** (0.119)	0.707*** (0.175) [0.324, 1.411]	0.404*** (0.0883) [0.044, 0.515]	0.482*** (0.0931) [0.252, 0.702]
First-stage robust F-stat				14.47		
Studies	18	18	18	18	18	18
Observations	167	167	167	167	167	167
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.450 (0.347)	P = 0.613 (0.177)	L = 0.188 (p = 0.911)
Effect beyond bias	0.446*** (0.072)	0.454*** (0.030)	0.349** (0.163)	0.405*** (0.028)	0.426*** (0.088)	0.427*** (0.103)
Studies	18	18	18	18	18	18
Observations	167	167	167	167	167	167
Part 2: Seven-Factor Model						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.142 (0.137) [-0.666, 0.555]	-0.0729 (0.0547)	0.305* (0.155)	0.624 (0.557) [NA] {0.0184, NA}	0.0683 (0.296) [-0.571, 0.948]	0.226 (0.265) [-0.732, 0.644]
Effect beyond bias (κ)	0.326*** (0.0392) [0.239, 0.413]	0.308*** (0.0141)	0.200** (0.0730)	0.128 (0.150) [NA]	0.284*** (0.0330) [0.132, 0.361]	0.222*** (0.0641) [0.073, 0.375]
First-stage robust F-stat				3.41		
Studies	33	33	33	33	33	33
Observations	298	298	298	298	298	298
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.019 (0.173)	P = 0.900 (0.212)	L = 0.269 (p = 0.874)
Effect beyond bias	0.229*** (0.036)	0.297*** (0.013)	0.325*** (0.059)	0.298*** (0.008)	0.302*** (0.059)	0.305*** (0.040)
Studies	33	33	33	33	33	33
Observations	298	298	298	298	298	298

Notes: Part 1: Sample in which the one-factor model or its modifications are used to estimate the alpha. Part 2: Sample in which the seven-factor model or its modifications are used to estimate the alpha. [NA] = confidence interval could not be bounded. $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

funds.

In Part 2 of Table 4, we report results of our tests of selective publication for the alpha coefficients based on the seven-factor model. Similarly to the results on the one-factor model reported in Part 1 of Table 4, the results in Part 2 of Table 4 show little evidence of publication selection bias. The λ coefficients are insignificant with the exception of the between effect esti-

mation (BE) that produces marginally significant λ of 0.305 (SE 0.155, significant at 10% level). Furthermore, the κ coefficients that reflect the expected value alpha coefficients after adjusting for selective publication range from 0.128 to 0.326, which is lower than the corresponding range in Part 1 of Table 4. The range is also below the unconditional mean of all monthly alpha estimates in our sample of 0.36. This magnitude of the κ coefficients is unlikely to be driven by selective publication.

Taken together, our results do not suggest that the heterogeneity in methodological approaches used for estimating alpha coefficients reported in primary studies is the underlying reason for not detecting any publication bias. We do not observe a significant publication bias even when concentrating on fairly homogeneous subsamples of alphas that are estimated using one of the common risk models.

5.3 Instrumental Variables

We further consider a subsample of alphas estimated based on IV, for which selective publication is particularly likely. Prior research shows that IV-based estimates tend to suffer from publication bias more frequently than estimates based on other techniques (Brodeur *et al.*, 2020). The authors argue that research methods that offer researchers more degrees of freedom are more likely to suffer from selective publication as researchers may exercise discretion in choosing research designs that help them achieve results that may be viewed as more attractive for publication. The choice of an IV and the specific way of measuring it give researchers considerable leeway. Researchers may choose to report IV-based estimates that are consistent with their prior beliefs or that are otherwise more attractive for publication. Brodeur *et al.* (2020) show that when IVs are relatively weak, the second stage results are likely to be close to the conventional thresholds for statistical significance, which is consistent with selectivity in the process that determines what coefficients eventually get published.

Motivated by this argument recently proposed in research literature we test for selective publication within the subsample of IV-based alpha coefficients. Primary studies typically use higher moments of the distribution of returns, such as skewness and kurtosis, as IV for the excess returns of the mimicking portfolios. This approach follows earlier research that shows that higher moments of the returns distribution are valid instruments and they are effective

in removing errors-in-variables problem (Durbin, 1954; Pal, 1980; Dagenais & Dagenais, 1997). We collect 46 IV-based alpha estimates from 3 different studies.

Our results reported in Part 1 of Table 5 are consistent with the proposition in prior literature that IV-based estimates tend to exhibit a greater publication selection bias. Five out of seven λ coefficients are positive and statistically significant at 5% level or better. The positive association between reported alphas and their standard errors indicates that highly positive alpha estimates tend to be reported when they are rather imprecise, i.e. they have a large standard error. Such a pattern is characteristic of selective publication. Furthermore, we also observe that for the subsample of IV-based estimates the magnitude of the κ coefficients that represent the expected value of alpha estimates after adjusting for selective publication is substantially lower than in our main results. The κ coefficients reported in Part 1 of Table 5 range from -0.411 to 0.298, many of them are close to 0, and five out of twelve are actually negative. This suggests that after correcting the IV-based alpha estimates for selective publication there is only limited evidence that they actually are positive and statistically significant. In fact, in contrast to our previous results, only two out of twelve κ coefficients are statistically different from zero. Consistent with the *a priori* expectations this evidence suggests that the composition of the pool of published IV-based alpha estimates tends to be affected by selective publication. These findings thus provide one of the first pieces of out-of-sample evidence in support of the recent proposition that IV-based estimates are more likely to suffer from publication bias than estimates based on other techniques Brodeur *et al.* (2020).

In contrast, our results based on the subsample of the remaining alpha coefficients that are not estimated with the use of IV reported in Part 2 of Table 5 are in line with our main results. All λ coefficients are statistically insignificant, which indicates that these alpha estimates are not substantially affected by the publication selection bias. In comparison to the full-sample results, within this subsample even the λ coefficient based on the Kinked-meta model is statistically insignificant. Furthermore, the κ coefficients fall within a fairly narrow range between 0.276 and 0.388, which is very similar to the full sample result. Taken together, the subsample of alpha coefficients that are not estimated based on IV do not seem to be affected by publication bias and their mean value corrected for any (small) biases are quite close to the unconditional sample mean of 0.36.

Table 5: Instrumental variables

Part 1: Methods Using Instrumental Variables						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	1.378*** (0.221) [0.760, 1.634]	1.307** (0.283)	3.514 (6.083)	2.459 (2.111) [NA] {-1.512, 6.430}	2.418*** (0.299) [1.886, 2.945]	1.445*** (0.178) [0.776, 1.647]
Effect beyond bias (κ)	0.127 (0.0851) [-0.102, 0.235]	0.144 (0.0691)	-0.411 (1.525)	-0.137 (0.438) [NA]	-0.0601 (0.0875) [-0.195, 0.260]	0.104 (0.102) [-0.113, 0.267]
First-stage robust F-stat				155.41		
Studies	3	3	3	3	3	3
Observations	46	46	46	46	46	46
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				2.418*** (0.431)	P = 0.341 (0.103)	L = 3.551 (p = 0.169)
Effect beyond bias	-0.036 (0.027)	0.018 (0.067)	0.078 (0.091)	-0.060 (0.048)	0.231*** (0.088)	0.298*** (0.071)
Studies	3	3	3	3	3	3
Observations	46	46	46	46	46	46
Part 2: Methods Not Using Instrumental Variables						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.0335 (0.191) [-0.543, 0.442]	-0.0230 (0.220)	0.0147 (0.132)	0.203 (0.351) [-0.493, 0.999] {-0.597, 1.003}	0.284 (0.321) [-0.453, 1.080]	0.0364 (0.131) [-0.351, 0.458]
Effect beyond bias (κ)	0.366*** (0.0429) [0.277, 0.459]	0.363*** (0.0553)	0.358*** (0.0487)	0.305*** (0.0849) [0.138, 0.469]	0.308*** (0.0440) [0.192, 0.424]	0.353*** (0.0382) [0.271, 0.435]
First-stage robust F-stat				12.98		
Studies	74	74	74	73	74	74
Observations	973	973	973	933	973	973
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.166 (0.108)	P = 0.636 (0.100)	L = 0.378 (p = 0.828)
Effect beyond bias	0.333*** (0.026)	0.329*** (0.009)	0.355*** (0.092)	0.324*** (0.008)	0.276*** (0.030)	0.388*** (0.045)
Studies	74	74	74	74	74	74
Observations	973	973	973	973	973	973

Notes: Part 1: Sample where the instrumental variable approach (including 2SLS, GMM, Hasuman) is used for estimation of the alpha. Part 2: Sample where the instrumental variable approach is not used for estimation of the alpha (mostly ordinary least squares). [NA] = confidence interval could not be bounded. $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Top Journals

In this section we report our results for the subsample of alpha estimates published in the Top 3 (the *Journal of Finance*, the *Journal of Financial Economics*, the *Review of Financial Studies*), or the Top 5 (plus the *Journal of Financial and Quantitative Analysis*, and the *Review of*

Finance) finance journals. These journals are particularly prominent in the field of finance, and so the alpha estimates they publish are likely to be particularly impactful in shaping researchers' views of hedge funds performance. In addition, many researchers are strongly incentivized to publish their research in these leading journals. This implies that if publication of hedge fund performance is selective, the bias is likely to be particularly strong in these journal.

Table 6 shows our results for the subsamples of alphas collected from the Top 3 (Part 1) and Top 5 (Part 2) finance journals. We observe that these results are mostly consistent with our main findings reported in Table 2. In line with our main results, we find little evidence of publication selection bias for the alpha estimates published in top finance journals. Most of the λ coefficients that capture the effect of selective publication are statistically insignificant. Furthermore, some of the λ coefficients are positive and others are negative, which points towards the absence of a systematic tendency to over-report or under-report high estimates of hedge fund performance. Only two λ coefficients are statistically significant at the conventional 5% level. In Part 1, in the model where the observations are weighted by the inverse of the number of estimates per study (wNOBS) the λ coefficient is positive and significant. In contrast, Part 2 shows a significant negative λ coefficient for the model that includes study-level fixed effects (FE). Thus, the only two statistically significant results point in the opposite direction. Consistent with our main results, we conclude that also our results based on subsamples of alpha estimates published in the Top 3 and Top 5 finance journals exhibit little signs of publication selection bias.

We also observe the magnitude of the κ estimates that reflect the estimated magnitude of the monthly alphas adjusted for the publication selection bias. These results are again broadly consistent with our main findings based on the full sample. All the κ coefficients fall within a fairly narrow interval of (0.265, 0.358) for the Top 3 journals and (0.263, 0.355) for the Top 5 journals. These intervals for the “corrected” alphas published in the top finance journals greatly overlap with the corresponding interval that we observe for the full sample (0.274, 0.386). In fact, the intervals for the top finance journals are slightly narrower which may be driven by greater consistency in the data sets and estimation methods that may be required by these leading journals. Thus, we again conclude that our results for the Top 3 and Top 5 finance journals do not materially differ from our main findings based on the full sample.

Table 6: Top journals

Part 1: Top 3 journals						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	0.00308 (0.104) [-0.226, 0.388]	-0.0876* (0.0469)	0.125 (0.0982)	0.0711 (0.190) [NA] {-0.301, 0.443}	0.241 (0.441) [-1.055, 2.069]	0.131*** (0.0363) [-0.177, 0.594]
Effect beyond bias (κ)	0.328*** (0.027) [0.265, 0.416]	0.352*** (0.012)	0.358*** (0.049)	0.310*** (0.057) [NA]	0.274*** (0.059) [0.086, 0.386]	0.356*** (0.044) [0.253, 0.450]
First-stage robust F-stat				2.444		
Studies	16	16	16	16	16	16
Observations	218	218	218	218	218	218
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				0.113 (0.257)	P = 0.817 (0.188)	L = 0.289 (p = 0.866)
Effect beyond bias	0.265*** (0.031)	0.314*** (0.014)	0.353*** (0.055)	0.314*** (0.008)	0.272*** (0.070)	0.306*** (0.070)
Studies	16	16	16	16	16	16
Observations	218	218	218	218	218	218
Part 2: Top 5 journals						
<i>Panel A: linear</i>	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.0537 (0.114) [-1.000, 0.182]	-0.121*** (0.040)	0.0916 (0.167)	-0.00437 (0.237) [NA] {-0.469, 0.460}	0.0795 (0.413) [-1.003, 1.686]	0.0893 (0.060) [-0.517, 0.227]
Effect beyond bias (κ)	0.300*** (0.042) [0.187, 0.397]	0.317*** (0.011)	0.354*** (0.074)	0.287*** (0.075) [NA]	0.279*** (0.053) [0.114, 0.384]	0.355*** (0.060) [0.234, 0.481]
First-stage robust F-stat				2.10		
Studies	22	22	22	22	22	22
Observations	256	256	256	256	256	256
<i>Panel B: non-linear</i>	Top10	WAAP	Stem-based	Kinked-meta	Selection model	p-uniform*
Publication bias				-0.071 (0.230)	P = 0.956 (0.216)	L = 0.307 (p = 0.858)
Effect beyond bias	0.263*** (0.030)	0.312*** (0.012)	0.355*** (0.061)	0.313*** (0.008)	0.289*** (0.061)	0.343*** (0.064)
Studies	22	22	22	22	22	22
Observations	256	256	256	256	256	256

Notes: Part 1: Sample extracted from top 5 journals in finance (*Journal of Finance*, *Journal of Financial Economics*, *Review of Financial Studies*, *Review of Finance*, *Journal of Financial and Quantitative Analysis*). Part 2: Sample extracted from top 3 journals in finance (*Journal of Finance*, *Journal of Financial Economics*, *Review of Financial Studies*). [NA] = confidence interval could not be bounded. $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5 Overview

Table 7 provides an overview of our subsample results. The table shows that, for the full sample, our estimates based on various techniques of the “representative” alpha coefficient corrected for the publication selection bias range from 0.274 and 0.386. The mean and median values of

0.335 and 0.338 respectively are both fairly close to the unconditional sample mean of 0.36. Relative to these full sample results, our estimates of the “representative” alpha have a lower minimum of 0.194 for the subsample of alphas treated the survivorship and/or backfilling biases, and a higher upper bound for the subsample of alphas untreated for either of the biases. The range of “corrected” alphas is wider and higher for alphas estimated with the use of the one-factor model (0.349, 0.707) relative to the seven-factor model (0.128, 0.326). While the range of the “representative” alpha coefficients estimated without the use of IV (0.276, 0.388) virtually co-incides with our full sample results, the range of the “corrected” alpha estimates based on IV techniques (-0.411, 0.298) is substantially wider and includes negative values. Finally, the ranges of the “corrected” alpha estimates based on observations published in the Top 3 (0.265, 0.358) and Top 5 (0.263, 0.355) are only slightly below our full sample results.

Table 7: Results overview

Table	Part	Note	#Studies	#Alphas	Mean	StDev	Min	Md	Max
Table 2		Full sample	74	1019	0.335	0.033	0.274	0.338	0.386
Table 3	1	Bias treated	50	605	0.273	0.048	0.194	0.265	0.351
Table 3	2	Bias untreated	29	414	0.390	0.077	0.282	0.380	0.521
Table 4	1	1F model	18	167	0.467	0.095	0.349	0.437	0.707
Table 4	2	7F model	33	298	0.269	0.061	0.128	0.298	0.326
Table 5	1	Instruments	3	46	0.025	0.187	-0.411	0.048	0.298
Table 5	2	No instruments	74	973	0.338	0.032	0.276	0.343	0.388
Table 6	1	Top 3	16	118	0.317	0.034	0.265	0.314	0.358
Table 6	2	Top 5	22	256	0.314	0.032	0.263	0.313	0.355

Notes: The table provides an overview of the results presented in this paper. *Table* and *Part* specify the table and its part where the results are reported. *Note* provides a brief description of a given set of results. *#Studies* and *#Alphas* show the number of studies and the number of alpha estimates a given set of results is based on. *Mean*, *StDev*, *Min*, *Md*, and *Max* refer to the mean value, the standard deviation, the minimum, the median value, and the maximum of a given set of results.

Our results make several important contributions to prior literature. First, we synthesize fragmented empirical evidence on hedge fund performance and present estimates that are corrected for any publication selection bias. Second, our results demonstrate that despite the prevalence of the publication selection bias in numerous other research settings, publication may not be selective when there is no strong *a priori* theoretical prediction about the sign of estimated coefficients, which may induce greater readiness to publish statistically insignificant results. Third, we provide one of the first out-of-sample tests of the proposition by (Brodeur *et al.*, 2020) who argues that IV-based estimates tend to suffer from publication bias more

frequently than estimates based on other techniques.

6 Conclusion

We perform a meta-analysis of prior empirical studies evaluating hedge fund performance. We examine whether published estimates of hedge fund alphas are affected by publication selection bias and by data biases. We observe that, prior research detects publication selection bias in a wide range of economic and finance settings, e.g. Stanley (2001, 2005); Stanley & Doucouliagos (2010); Havranek (2015); Brodeur *et al.* (2016); Bruns & Ioannidis (2016); Stanley & Doucouliagos (2017); Christensen & Miguel (2018); Brodeur *et al.* (2020); Blanco-Perez & Brodeur (2020), and Zigravova *et al.* (2021). In contrast to these findings, using a wide range of techniques and data partitions we do not detect selective publication in hedge fund performance literature with the exception of for estimates based on IV. In contrast, we provide evidence that not controlling for the potential biases in the underlying data (e.g. the backfilling bias and the survivorship bias) affects reported alpha coefficients.

Fragmentation of hedge fund performance data and the wide range of alternative approaches for controlling for risk give researchers considerable discretion over the design of their research. This potentially creates opportunities for selective publication because the use of various estimation techniques based on different data sources may yield diverse results, some of which may be more attractive for publication than others. Our results demonstrate that despite the prevalence of the publication selection bias in numerous other research settings, publication may not be selective when there is no strong *a priori* theoretical prediction about the sign of estimated coefficients, which may induce greater readiness to publish statistically insignificant results.

The heterogeneity in methodological approaches and data sources used in estimating hedge funds' alphas opens up additional research opportunities. Future research can examine whether and how the various aspects of methodological choices affect the magnitude of reported alpha coefficients. Our aim in this paper is to propose a representative alpha coefficient that is aggregated across the plurality of these approaches and corrected for the publication and data biases. Therefore, in this study we provide robustness checks based on subsamples that narrow down the pool of collected alpha estimates to more homogeneous subsets but we do not explicitly

exploit the full sample heterogeneity to analyze and draw conclusions about individual subsets or about the relative magnitude of alpha coefficients for the individual subsets. We leave the analysis of the impact of this heterogeneity on the reported alpha coefficients for future research that can examine the importance of various dimensions of methodological choices on the alpha coefficients reported in primary studies.

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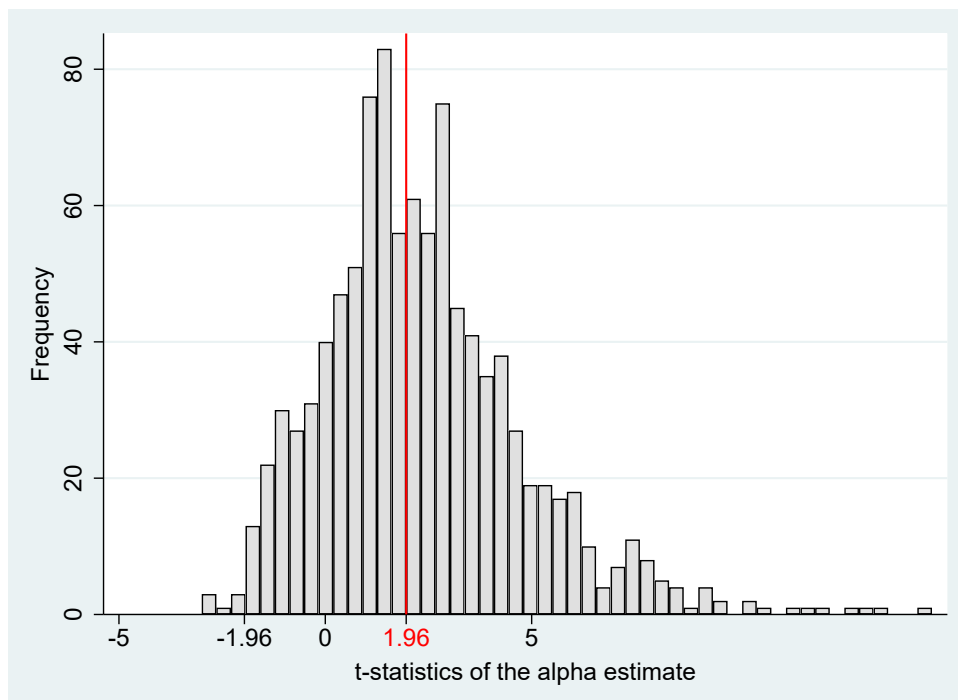
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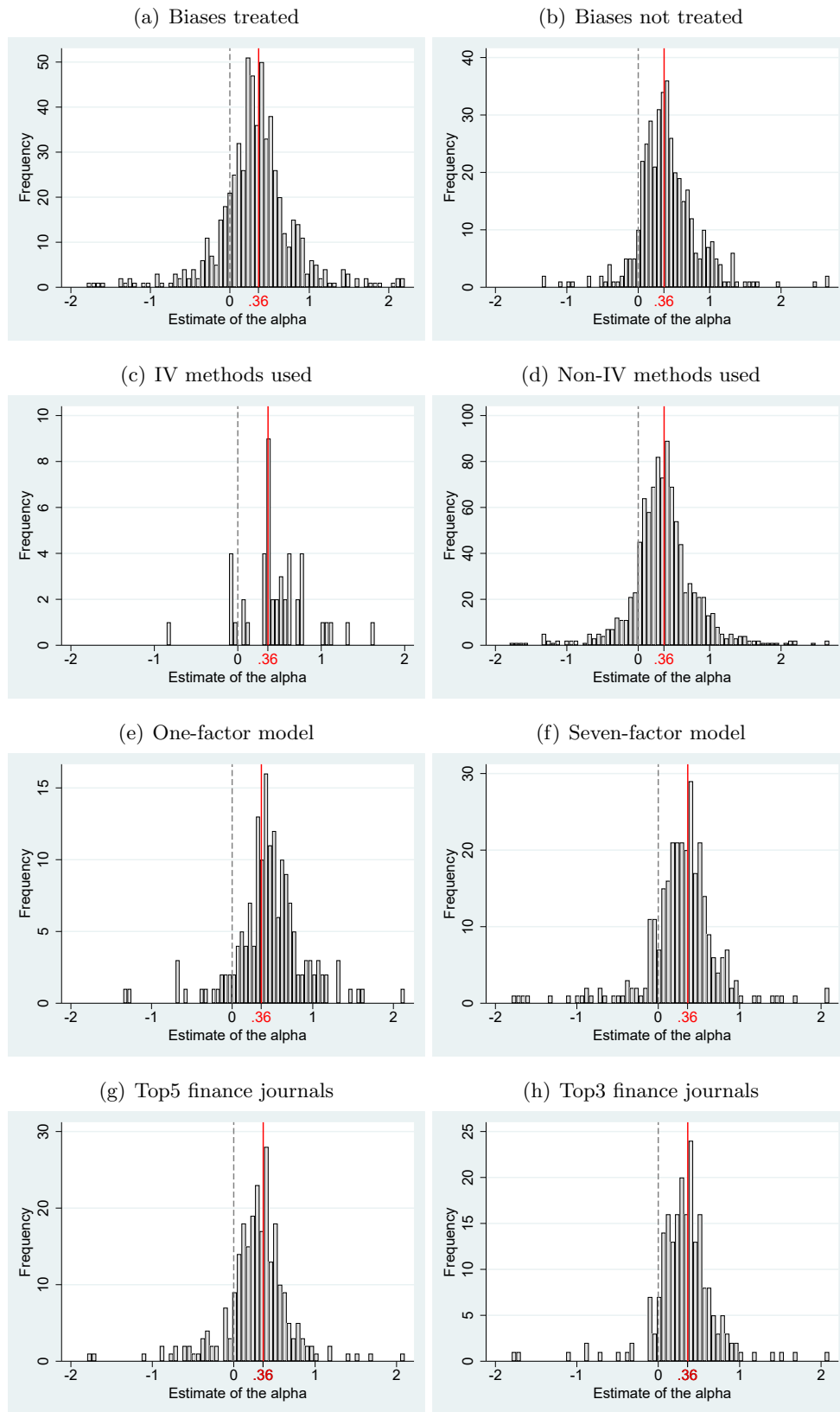
Appendix

Figure A1: Distribution of t -statistics



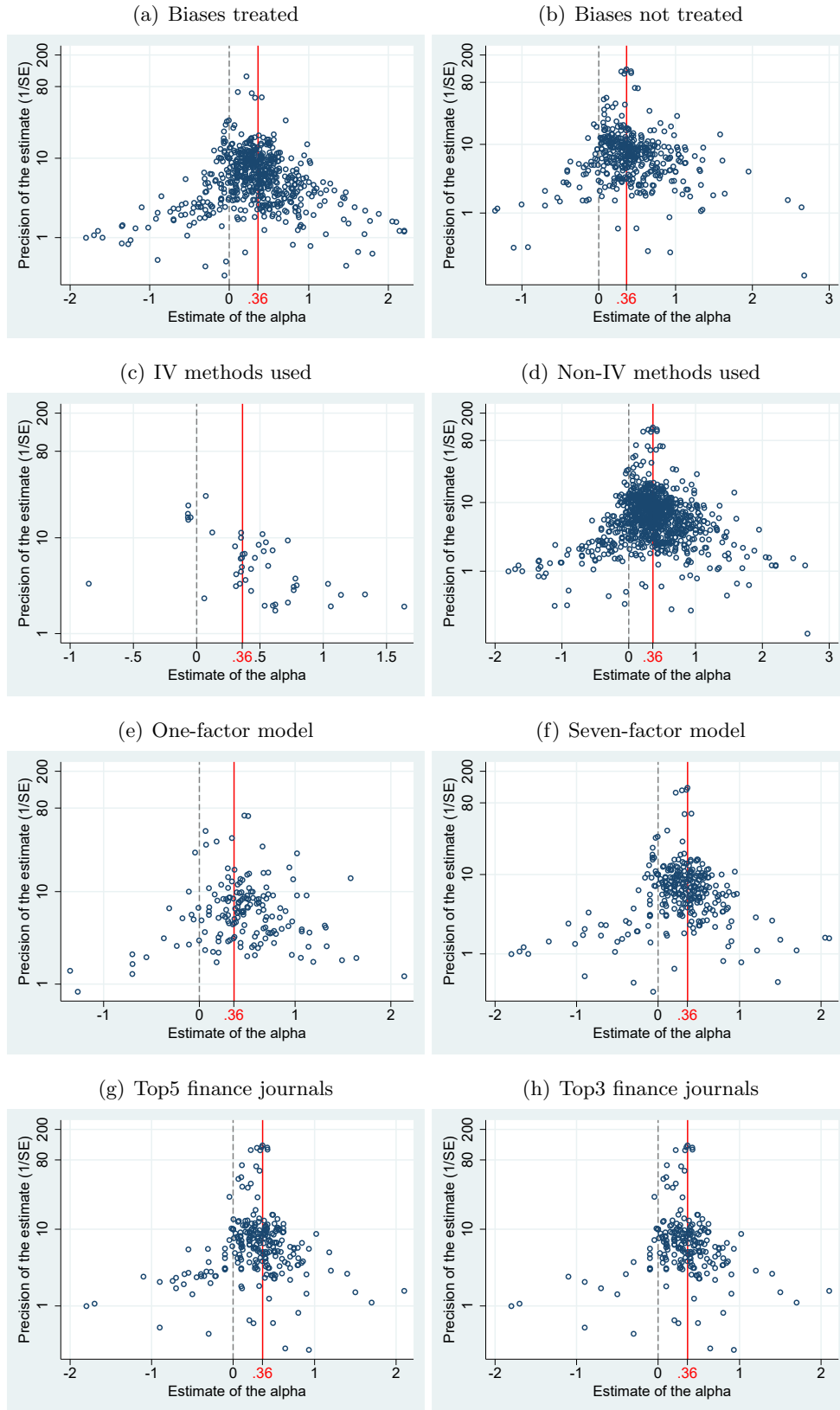
Note: The figure represents the distribution of t -statistics of the reported estimates of the alpha. Red lines represents critical value of 1.96 associated with significance at the 5% level and the value of 0 associated with changing the sign of the estimate. We exclude estimates with large t -statistics from the figure for ease of exposition but include them in statistical tests.

Figure A2: Histograms for subsamples



Notes: The figure depicts funnel plots of the reported alphas divided based on the treatment of biases, the implementation of methods (IV = instrumental variables) and models, and the quality of journals in finance. The solid vertical line denotes the sample mean; the dashed vertical line denotes the null alpha.

Figure A3: Funnel plots for subsamples



Notes: The figure depicts funnel plots of the reported alphas divided based on the treatment of biases, the implementation of methods (IV = instrumental variables) and models, and the quality of journals in finance. The solid vertical line denotes the sample mean; the dashed vertical line denotes the null alpha.

Table A1: Specification test of Andrews & Kasy (2019)

	All estimates	Bias treated	Bias not treated	IV estimates	Non-IV estimates
Correlation	0.330 [0.264, 0.392]	0.359 [0.275, 0.430]	0.318 [0.191, 0.431]	0.389 [-0.031, 0.769]	0.322 [0.243, 0.379]
Observations	1,019	605	414	46	973
		One-factor model	Seven-factor model	Top 5 journals	Top 3 journals
Correlation		0.096 [-0.091, 0.255]	0.287 [0.166, 0.407]	0.262 [0.155, 0.375]	0.262 [0.155, 0.375]
Observations		167	298	256	218

Notes: The table shows the inverse publication-probability-weighted correlations between $\log(\alpha)$ and $\log(SE(\alpha))$, tests developed by Kranz & Putz (2022) for viability of Andrews & Kasy (2019) publication bias test. If all the assumptions of the selection model hold, the correlation should be zero. Bootstrapped standard errors in parentheses.

Table A2: Clustered for author teams

	OLS	FE	BE	IV	WLS	wNOBS
Publication bias (λ)	-0.0152 (0.171) [-0.488, 0.365]	-0.0265 (0.206)	0.0602 (0.131)	0.178 (0.359) [-0.617, 1.114] {-0.639, 1.067}	0.324 (0.310) [-0.377, 1.097]	0.0497 (0.124) [-0.313, 0.458]
Effect beyond bias (κ)	0.366*** (0.0453) [0.265, 0.470]	0.369*** (0.0516)	0.350*** (0.0474)	0.316*** (0.0852) [0.150, 0.474]	0.301*** (0.0440) [0.183, 0.416]	0.353*** (0.0397) [0.274, 0.434]
First-stage robust F-stat				11.71		
Studies	74	74	74	73	74	74
Observations	1,019	1,019	1,019	979	1,019	1,019

Notes: The table reports the results of the regression $\alpha_{ij} = \kappa + \lambda \cdot SE(\alpha_{ij}) + \epsilon_{ij}$, where α_{ij} denotes the i -th alpha coefficient estimated in the j -th study, and $SE(\alpha_{ij})$ denotes its standard error. FE: study-level fixed effects, BE: study-level between effects, IV: the inverse of the square root of the number of observations is used as an instrument for the standard error, WLS: model is weighted by the inverse of the standard error of an estimate, wNOBS: model is weighted by the inverse of the number of estimates per study. Standard errors, clustered at the level of authors, are reported in parentheses. 95% confidence intervals from wild bootstrap in square brackets (Roodman *et al.*, 2018). In curly brackets we show the two-step weak-instrument-robust 95% confidence interval based on Andrews (2018) and Sun (2018). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Tests of p -hacking

	20 bins	15 bins	10 bins
Test for non-increasingness	0.469	0.179	0.403
Test for monotonicity and bounds	0.242	0.223	0.481
Observations ($p \leq 0.15$)	663	663	663
Total observations	1,019	1,019	1,019

Notes: Results of p -hacking tests based on Elliott *et al.* (2022) for the whole sample.